



December 9, 2025

Madison County VA Clinic

Agency Specific Requirements (ASR)

1. **INTRODUCTION**
 2. **GENERAL REQUIREMENTS**
 - 2.1 General
 - 2.2 Codes
 - 2.3 Standards
 - 2.4 References
 - 2.5 Design, Construction, and post award activities
 3. **PLANNING AND DESIGN CRITERIA**
 - 3.1 Background
 - 3.2 PACT Guiding Principles
 - 3.3 Community Based Outpatient Clinics (CBOC) Planning Approach
 - 3.4 Room Data Matrix
 - 3.5 Surgical Design Criteria
 4. **TECHNICAL NARRATIVE FOR CONSTRUCTION**
 - 4.1 Master Construction Specifications
 - 4.2 Site/Civil
 - 4.3 Architecture
 - 4.4 Interior Design
 - 4.5 Mechanical
 - 4.6 Plumbing
 - 4.7 Electrical
 - 4.8 Lighting
 - 4.9 Telecommunications
 5. **Lessor-Provided Services**
 - 5.1 Maintenance
 - 5.2 Utilities
 - 5.3 Services
 6. **Government-Provided Services**

Appendix A: Room Data Matrix

Appendix B: Hardware Groups and Modifiers

Appendix C: Janitorial Services
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1. INTRODUCTION

- 1.1. The purpose of this document is to consolidate the governing codes, standards, references, and guidelines to provide performance-based standards for this VA leased CBOC.
- 1.2. The governing codes, standards, and references are based on a VA affiliated outpatient clinic.
- 1.3. The design of the VA leased CBOC is based on the Prototype for Standardized Design and Construction of Community Based Outpatient Clinics which is described in Part 3 Planning and Design Criteria.
- 1.4. Commonly Used Acronyms and Definitions:
 - 1.4.1. PACT: Patient Aligned Care Team. Provides service to veterans that is patient-centered with the right care at the right time by the right person. A typical “teamlet” serving a veteran consists of four caregivers: A Primary Care Provider – physician (MD), physician assistant (PA) or nurse practitioner (NP); a Registered Nurse (RN); a clinical staff assistant (LPN or Medical Assistant) and an administrative staff member (Clerk).
 - 1.4.2. CBOC: Community Based Outpatient Clinic. Is VA-operated, VA-funded, or VA-reimbursed site of care which is located separate from a VA Medical Facility. A CBOC can provide primary, specialty, subspecialty, mental health, or any combination of healthcare delivery services that can be appropriately provided in an outpatient setting.
 - 1.4.3. PFD: Program for Design. Net square feet (NSF) listing all spaces and rooms that are to be included in a construction project.
 - 1.4.4. VA: Veterans Affairs
 - 1.4.5. VHA: Veterans Health Administration

2. GENERAL REQUIREMENTS

- 2.1. General
 - 2.1.1. The codes, standards, and references listed below indicate minimum performance requirements. Based on input from Owner representatives and recommendations from the architectural/engineering firm, minimum requirements or standards may be exceeded by site specific project design.
 - 2.1.2. Compliance is required with applicable codes and standards throughout the process of design, construction, acceptance, and on-going maintenance of the facility.
 - 2.1.3. Design and construction of the CBOC shall be in compliance with the requirements of the GSA Form L100 Global Lease and the codes, standards, and references listed below. This Leased CBOC Design Narrative covers construction materials and standards not fully addressed by the codes, standards, and references below.
 - 2.1.4. Use the most current edition at the date and time of bid submission
- 2.2. Codes
 - 2.2.1. All VA leased CBOC facilities are to conform to the most recent applicable codes, which include but is not limited to following:
 - 2.2.1.1. National Fire Protection Association (NFPA) 70 National Electric Code, NFPA 99 Health Care Facilities Code and all standards referenced therein, NFPA 101 Life Safety Code, current editions.
 - 2.2.1.2. Architectural Barriers Act Accessibility Standards (ABAAS), current edition.

Madison County VA Clinic Agency Specific Requirements

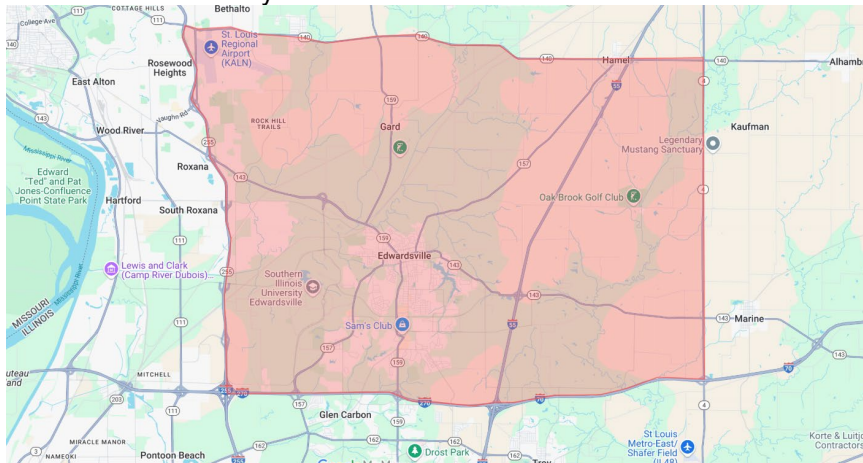
- 2.2.1.3. International Building Codes (IBC): 2018 or latest edition (within 2 years of release). The latest edition shall be used for structural and seismic design.
 - 2.2.1.3.1. International Building Code
 - 2.2.1.3.2. International Energy Code
 - 2.2.1.3.3. International Mechanical Code
 - 2.2.1.3.4. International Plumbing Code
 - 2.2.1.4. State and local codes as required by the local Authority Having Jurisdiction (AHJ).
 - 2.2.1.5. For the Lessor to properly determine occupancy type, VA intends to simultaneously treat 0 patients at any given time who are incapable of self-preservation (i.e. Business Occupancy).
 - 2.2.1.6. Lessor shall maintain utilities and building equipment as prescribed by local codes, as outlined within this document.
- 2.3. Standards
- 2.3.1. All leased CBOC facilities are to conform to the following standards:
 - 2.3.1.1. FGI Guidelines for the Design and Construction of Outpatient Facilities, current edition (referred to as FGI Guidelines herein).
 - 2.3.1.2. Facility Security Level (FSL) shall be as outlined in GSA Form L100 Global Lease.
 - 2.3.1.3. Sustainability standards shall be as outlined in GSA Form L100 Global Lease.
 - 2.3.1.4. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standard 170, Ventilation of Health Care Facilities, current edition (referred to as ASHRAE 170 herein)
 - 2.3.1.5. ASHRAE Standard 12, Minimizing the Risk of Legionellosis Associated with Building Water Systems, current edition.
 - 2.3.1.6. ASHRAE Standard 188, Legionellosis: Risk Management for Building Water Systems, current edition.
 - 2.3.1.7. ASHRAE Standard 62.1, Ventilation for Acceptable Indoor Air Quality, current edition (referred to as ASHRAE 62.1 herein)
 - 2.3.1.8. ANSI/ASHRAE/IES Standard 90.1-2019 -- Energy Standard for Buildings Except Low-Rise Residential Buildings
 - 2.3.1.9. Sheet Metal and Air Conditioning Contractors National Association (SMACNA) Standard, HVAC Duct Construction Standards - Metal and Flexible.
 - 2.3.1.10. Illuminating Engineering Society North America (IESNA) Handbook - latest edition
 - 2.3.1.11. VA Signage Design Manual PG 18-12
 - 2.3.1.12. [H 18-8 Seismic Design Handbook](#) Seismic Facility Criticality Designation (FCD), based on the Program for Design, is defined as: *Ancillary*
 - 2.3.1.13. VA Enterprise Facility IT Support Infrastructure Standard, most current.
- 2.4. References
- 2.4.1. The following guides, manuals, and other references developed by VA are for reference only, however, they are applicable to VA leased CBOCs:
 - 2.4.1.1. Prototype for Standardized Design and Construction of Community Based Outpatient Clinics <https://www.cfm.va.gov/til/prototypes.asp#CBOC>
 - 2.4.1.2. PACT Space Module Design Guide <https://www.cfm.va.gov/til/dGuide/dgPACT.pdf>
 - 2.4.1.3. Room Templates, Leased Community Based Outpatient Clinics <https://www.cfm.va.gov/til/rTemplate/LeasingCBOCW.asp>
 - 2.4.1.4. VA Design Guides (PG18-12)
- 2.5. Design, Construction, and post award activities schedule for completion of space.
- 2.5.1. At least four (4) design submissions (Level 1/Level 2 DIDs- 35%, 65%, 95% and 100% and CDs) shall be submitted by the Lessor, and approved in writing by VA, before proceeding to the next submission or construction. Lessor's A/E firm will facilitate design review meetings

Madison County VA Clinic Agency Specific Requirements

- within 14 calendar days of each submission. Design submissions should follow Design Submission Instructions found in Program Guide, PG-18-15, Volume C.
- 2.5.2. Not later than 30 days after the acceptance of the Space, the Lessor, at Lessor's expense, shall furnish to the Government a complete set of Computer Aided Design (CAD) and Building Information Models (BIM) files and PDFs of as-built floor plans showing the Space under Lease, as well as corridors, stairways, and core areas. The plans shall have been generated by at least 2020 CAD or 2020 Revit program. The required file extension is ".DWG," or ".RVT". Clean and purged BIM and PDF files shall be submitted in a digital format. They shall be labeled with Building name, address, list of drawing(s), date of the drawing(s), and Lessor's architect and architect's phone number. The Lessor's operator shall demonstrate the submission on VA equipment, if requested by the LCO.

3. PLANNING AND DESIGN CRITERIA

- 3.1. Background. VHA has committed to the development of a patient-centered medical home model for enhanced outpatient primary care services to veterans. The result was a patient care delivery model and a set of planning and design standards.
- 3.2. PACT Guiding Principles. An operational and planning design guide was developed which identified key principles and objectives. Refer to PACT Space Module Design Guide under References Section 2.4.
- 3.3. Community Based Outpatient Clinics (CBOC) Planning Approach. The development of the PACT Planning Module was complimented by the development of a Prototype for Standardized Design and Construction of Community Based Outpatient Clinics. Using the PACT Module as the building block, prototypes were developed for three types and sizes of outpatient facilities.
- 3.4. VA Project Specific Requirement. The Department of Veterans Affairs has a need for 17,000 (contiguous + circulation space) ABOA Square Feet (ANSI/BOMA= American National Standards Institute/Building Owners and Managers Association) not to exceed a maximum of 18,750 ABOA SF (including circulation) of clinical space for a Veterans Health Administration Outpatient Clinic in Edwardsville, IL. See the Program For Design (PFD) and Room Data Matrix for details.
- 3.5. Property Delineated Area. The property shall be within the delineated area and have municipal utilities available. Delineated Area is as follows:
- 3.5.1.1. Northern boundary: Highway 140
 - 3.5.1.2. Southern boundary: I-270/70
 - 3.5.1.3. Western boundary: I-255
 - 3.5.1.4. Eastern boundary: IL Route 4



3.6. Room Data Matrix

3.6.1. A collection of typical rooms within leased CBOCs. It lists the required finishes, STC ratings, ceiling heights, door size and type, hardware sets, HVAC space ventilation, number of electrical receptacles, lighting controls and fixture types, and number of standard density work area outlets for each room type.

3.6.2. The Lessor shall associate rooms listed on the program for design with the room data matrix by room name. Use department level room names and types when available, otherwise use general room/space for like room function.

3.6.3. FGI Equivalent for ASHRAE 170 – Where HVAC space ventilation requirements are omitted, ASHRAE 170 Tables 8-1 and 8-2 shall be followed. Column “FGI Equivalent for ASHRAE 170” lists the equivalent room name for use in the tables.

3.6.4. Additional Requirements in this Design Narrative take precedence over the standard in the Room Data Matrix.

3.6.5. The Room Data Matrix can be found in [Appendix A](#).

3.7. Space considerations.

3.7.1. Rooms other than storage rooms and corridors may not be over 3 times longer than they are wide. Treatment Exercise Area (PTEA1) should be more square than rectangular.

3.7.2. Area of building leased to VA must be roughly rectangular in shape and must not be more than 3 times longer than its width

4. TECHNICAL NARRATIVE FOR CONSTRUCTION

4.1. Master Construction Specifications for specific divisions and sections may be found at <https://vatilms.va.gov/vatilms/reports/PG-18-1>.

4.1.1. Building materials shall be asbestos-, mold- and lead-free.

4.1.2. Commissioning Service. Lessor will contract with an independent 3rd party commissioning agent (“Commissioning Agent”) to perform Acceptance and Testing Phase Commissioning in accordance with Chapter 8 of the VA’s Whole Building Commissioning Process Manual dated May 1, 2013 (<https://www.cfm.va.gov/til/Cx-RCx/CxManual.PDF>). Systems requiring commissioning include Fire Suppression, Plumbing, HVAC, Electrical, Communications, Electronic Safety and Security, and Integrated Systems Tests as further described in Chapter 3. The cost of the Commissioning shall be paid by the lessor and included in the build out cost.

4.2. Site/Civil

4.2.1. Site Consideration: Facility must be highly visible and located in a high traffic area accessible to and from main thoroughfares.

4.2.2. Pedestrian and Vehicle Access & Circulation

4.2.2.1. Minimum traffic lane width is 12 feet, and minimum sidewalk width is 4 feet. Curves for traffic lanes and radii at intersections must be adequately sized to prevent vehicles from encroaching on an opposing lane of traffic.

4.2.2.2. An accessible route must be provided from the public right-of-way abutting the site to the accessible building entrance.

4.2.2.3. Provide a service area with a loading dock designed to accommodate truck (WB-62 SU-30) maneuverability. Loading dock shall be 4 feet above the driveway. Platforms shall have a minimum depth of 8 feet front to back or between dock lift/leveler and back wall. Provide canopy over the platform with 14 feet of clearance from grade to the underside of the canopy. Canopy shall extend minimum of 4 feet beyond the edge of the dock for weather protection. Provide stair or ramp to the platform. Provide hydraulic dock levelers with 25,000 pounds capacity for recessed installation at loading dock.

- 4.2.2.4. Service area shall accommodate vehicles that pick-up trash and recycled materials. Locate service area away from public and patient areas.
- 4.2.2.5. Provide reflective traffic control signs as required for intersections, no parking lanes, and guidance of site traffic.
- 4.2.2.6. Public transportation must be available to the facility with the pickup/drop off located within 0.5 miles of the main building public entry point.
- 4.2.2.7. All concrete curbs must be painted high visible yellow and maintained as such. At minimum, all curbs shall be repainted every three years, and as needed.
- 4.2.2.8. Provide necessary curb cuts with truncated domes at all major pedestrian intersections with vehicle pathways.
- 4.2.2.9. During Construction, provide privacy screening on construction fencing to lessen social impacts.
- 4.2.3. Parking
 - 4.2.3.1. Parking lots with 90-degree stalls must have minimum parking stall dimensions of 9 feet by 18 feet, and a minimum drive aisle width of 24 feet. Angled parking must have one-way drive aisles with the same stall sizes as 90-degree parking. Angled parking drive aisle width must comply with a published design standard for a designated parking angle.
 - 4.2.3.2. Parking spaces should accommodate 100 cars. This includes 7 standard handicap spaces, 1 van-accessible handicap space, and the remaining parking spaces are open for staff or patients.
 - 4.2.3.3. Parking lots shall be resealed and striped every three years
- 4.2.4. Site Grading.
 - 4.2.4.1. Roads and walks should have a typical cross slope of 2% unless adequate surface drainage is provided by other slope conditions.
 - 4.2.4.2. Pedestrian pathways and ramps shall meet VA Barrier Free Design standard with maximum slope of 1:20. Ramps shall have rail on both sides.
 - 4.2.4.3. Exterior walks shall be 5ft in width.
 - 4.2.4.4. If site topography permits, site will be equipped with bushes and trees.
- 4.2.5. Site Flagpole: A flagpole will be provided and installed by the Lessor at a location to be approved by the LCO if the government is the sole occupant of the building. The flag will be provided by the Lessor, as part of shell rent and replaced at all times during the lease term when showing signs of wear. The flagpole will be 25 foot (minimum height). Flagpole shall include adequate lighting to illuminate flag between dusk and dawn. Flagpole shall be protected from vehicular traffic.
- 4.3. Architecture
 - 4.3.1. General Criteria. Refer to Appendix: Room Data Matrix for Door, Hardware and Acoustic requirements listed by Room Type.
 - 4.3.2. Accommodations for Care of Patients of Size. Refer to FGI Guidelines for Outpatient Facilities for accommodations for care of patients of size. Minimum requirements are to provide access to a Procedure Room and Procedure Room Toilet in one PACT Module using a 650lb capacity design requirement.
 - 4.3.3. Acoustics. Refer to FGI Guidelines for Outpatient Facilities for acoustics requirements for enclosed rooms. Refer to the Room Data Matrix-Interior Construction for information on rooms not covered under FGI or rooms with special acoustics requirements.
 - 4.3.3.1. To maximize sound privacy, STC ratings shall include the plenum/ceiling as well for Mental Health rooms and Conference rooms. If needed, walls shall extend to the deck above.
 - 4.3.3.2. Acoustical partitions are to be installed to provide visual privacy at multi-patient check-in counters as well as a designated "queue line" designation to provide an identifiable physical distance separation between the person checking in and the queue. The minimum distance from the front edge of the Check-in/Reception desk to the queue line shall be five (5) feet. Please refer

- to Design for Patient Privacy and Women Veterans' Health.
- 4.3.4. Equipment/Accessories/Furnishings
- 4.3.4.1. Refer to *Project Room Contents* tab of the Room Data Matrix. Items to be provided and/or installed by Lessor are noted on the list in Column K, *Acq Code* as "CC" or "VC." Provide partitions, partition backing, and above ceiling structural support as required for wall and/or ceiling mounted equipment, accessories, and furnishings. Provide required utility connections for scheduled equipment.
- 4.3.4.2. Lessor is responsible for providing all utilities to government furnished equipment.
- 4.3.4.3. Fire Extinguishers: Portable 10 lb. fire extinguishers will be provided, hung and maintained by the VA. Lessor is to install recessed fire extinguisher cabinets at appropriate locations to meet NFPA and AHJ.
- 4.3.4.4. Radiology Room. Provide Physicist calculations and certification of lead lining for specified radiological equipment (general X-ray).
- 4.3.5. Signage and Wayfinding.
- 4.3.5.1. Refer to VA Signage and Wayfinding Design Guide for signage standards and requirements at <https://vatilms.va.gov/vatilms/reports/PG-18-10> and *Design for Patient Privacy and Women Veterans' Health* at <https://vatilms.va.gov/vatilms/reports/Alerts/Design%20Alerts> for specific signage related to baby changing stations.
- 4.3.5.2. Marquee will be required if the property is located within a mall-type area. The Lessor will provide a lighted marquee and design.
- 4.3.5.3. During Construction, provide a 4 foot x 8 foot temporary sign visible from the public roadway with *Madison County VA Clinic Coming Soon*, and an appropriate rendering(s) of the building.
- 4.3.5.4. Post construction, Lessor shall register the location with Google Maps, Apple Maps, and Waze as *Madison County VA Clinic*.
- 4.3.6. Critical Dimensions
- 4.3.6.1. Patient-use corridors are to be 6 feet minimum. Staff corridors for transport of supplies from loading dock area to services shall be 8 feet minimum.
- 4.3.6.2. Refer to Room Data Matrix – Interior Construction for required ceiling heights by room type.
- 4.3.7. Entry Canopies
- 4.3.7.1. Provide non-combustible canopies over the following locations: patient entry to clinic, and receiving area/loading dock.
- 4.3.7.2. At patient entry, provide a covered patient drop-off zone with space for at least one full size passenger vehicle (19 feet long) and an accessible access aisle.
- 4.3.7.3. Provide clearance from grade to underside of canopy with for typical emergency and transport vehicles used in the area.
- 4.3.8. Elevators. Where elevators are provided in a leased facility, not less than one elevator car shall accommodate an ambulance stretcher of 24 inches by 84 inches in the horizontal open position and shall be identified by the international symbol for emergency medical services (star of life). The symbol shall be not less than 3 inches high placed inside on both sides of the hoist way door frame.
- 4.3.9. Slab Depressions and Thickened Slabs. At new construction, provide slab depression and thickened slabs as required to provide flush threshold installation at audiology booths, and other equipment requiring recessed installation or additional structure as recommended by equipment manufacturer. At these equipment locations, concrete slab levelness shall be level and true to meet vendor requirements and shall have a minimum flatness and levelness of FF=45 and FL=35. At existing construction, provide ramp to meet ABAAS.
- 4.3.10. Millwork
- 4.3.10.1. Quality compliance to conform to Architectural Woodwork Institute (AWI)

- and Woodwork Institute (WI) quality standards.
- 4.3.10.2. Wood Veneer Casework Construction: Combination of custom wood casework and countertops to conform to program and functional requirements. Location: Lobby, Reception. Construction: Flush overlay construction with concealed hinges. AWI custom grade.
- 4.3.10.3. Plastic Laminate Casework Construction: Custom plastic laminate casework to conform to program and functional requirements. Location: Typical.
 - 4.3.10.3.1. Construction: Flush overlay construction with concealed hinges, AWI Custom Grade.
 - 4.3.10.3.2. Cabinet Hardware:
 - 4.3.10.3.2.1. Concealed hinges, all metal, 110 degree opening, self-closing.
 - 4.3.10.3.2.2. Full extension drawer guides, 100 lbs.capacity.
 - 4.3.10.3.2.3. Door and drawer pulls: 4" center to center wire pull, 5/16" diameter round, 1 1/4" projection, satin chrome.
- 4.3.10.4. Solid Surface Fabrications: Hard, solid, non-porous mineral-filled acrylic resin (methyl methacrylate) material in color to be selected by designer. Thickness: 1/2" minimum.
 - 4.3.10.4.1. Locations:
 - 4.3.10.4.1.1. Transaction tops at nurse stations and reception desks.
 - 4.3.10.4.1.2. All counters in patient treatment areas to have solid surface countertop with 4" high minimum integral coved side and backsplash.
 - 4.3.10.4.1.3. Where scheduled as wall protection, provide 1/4" thick material with welded seams.
- 4.3.10.5. Heavy gauge stainless steel (SS): fully welded with a work surface load capacity of at least 300 lbs. per linear foot and antimicrobial. Locations: Preparations and Assembly (Utility Room, Soiled).
- 4.3.11. Joint Sealant Systems. Joints at plumbing fixtures, ceramic tile to be mildew resistant silicone.
- 4.3.12. Doors and Frames
 - 4.3.12.1. Refer to Room Data Matrix for door and hardware requirements by room type.
 - 4.3.12.2. Door details located at: <https://www.cfm.va.gov/til/rTemplate/sDetail-SD080000-01-02.pdf>
 - 4.3.12.3. Finish: Prefinished in compliance with AWI Section 1500.
 - 4.3.12.4. Automatic Sliding Doors: Single Slide Automatic Sliding Door, Class 1 Clean Room Certified, narrow stile, 84" wide w/ 35.3" nominal clear door opening, trackless, clear anodized finish, and touchless actuators.
 - 4.3.12.4.1. Comply with BHMA Standard ANSI A156.10 (BHMA 1601), Power Operated Pedestrian Door Standard and UL Standard UL 325, Electric Door, Drapery, Gate, Louver and Window Operators and Systems.
 - 4.3.12.4.2. Service Life: Provide automatic sliding doors capable of operating without failure of any component, for not less than 300,000 open and close cycles, with normal maintenance as defined in manufacturer's standard operating manual.
 - 4.3.12.4.3. Location: main entrance vestibule, two sets for air lock.
 - 4.3.12.4.4. In the event of negative air pressure conditions, provisions shall be made for equalizing air pressure.
 - 4.3.12.4.5. The Lessor shall provide permanent entryway systems (such as walk off mats) to control dirt and particulates from entering the building at all primary exterior entryways.
 - 4.3.12.5. Access Panels:
 - 4.3.12.5.1. Flush stainless steel at toilet areas.
 - 4.3.12.5.2. Flush gypsum board surface with concealed hinges at public and office areas.

- 4.3.12.5.3. Fire rated steel at rated construction.
- 4.3.13. Hardware
 - 4.3.13.1. General Note
 - 4.3.13.1.1. Hardware sets shall be provided as per Section 5 Hardware Groups and Modifiers and the Room Data Matrix. Provide extra heavy duty, Grade 1 hardware for all components.
 - 4.3.13.2. Cylinders and Keying: the keying system shall be an intelligent key system compatible with VASTLHCS keying system:
 - 4.3.13.2.1. Electronic Cylinders shall fit and function with LFIC locking hardware.
 - 4.3.13.2.2. Electronic Keys shall operate Electronic Cylinders.
 - 4.3.13.2.3. Electronic Cylinders and Electronic Keys shall be operated on VA TRM approved software, as listed on OIT approval site:
<https://www.oit.va.gov/Services/TRM/SearchPage.aspx>
 - 4.3.13.3. Electronic Cylinders
 - 4.3.13.3.1. Electronic cylinders shall eliminate the need to "rekey" mechanical cylinders, and allow electronic reprogramming of cylinders by authorized programming key.
 - 4.3.13.3.2. Electronic cylinder shall have stainless steel plug face and shell to resist environmental conditions
 - 4.3.13.3.3. Electronic cylinder shall have non-volatile memory to store 2,000 audit events, with the oldest record self-deleting after 2,000 events
 - 4.3.13.3.4. Electronic cylinder shall record unauthorized entry attempts from keys that are not in the system, are outside of the schedule date or time range, are not programmed to open the specific cylinder, or are stored on internal black list.
 - 4.3.13.3.5. Electronic cylinders shall store up to 40 blacklisted keys (keys lost or out of service)
 - 4.3.13.3.6. Electronic cylinders shall have drill resistant inserts to improve drill resistance
 - 4.3.13.3.7. Electronic cylinders shall have break-away plug face that shall fail secure in the event of over-torquing
 - 4.3.13.3.8. Electronic cylinders shall be independently tested to meet dust and water jet IP-68 and certified to specification CEI/IEC 60529, Edition 2.1
 - 4.3.13.3.9. Electronic cylinders shall be independently tested to meet dust and water jet IP-55 and certified to Specification CEI/IEC 60259 Edition 2.1 and DIN EN 1303, ICS 91.190
 - 4.3.13.3.10. Electronic cylinders shall be independently tested to meet salt fog testing for greater than 96 hours, certified to specification ASTM-B-117 and DIN EN 1303, ICS 91.190
 - 4.3.13.3.11. Electronic cylinders shall be independently tested to meet MIL-STD-202G, Method 201A, 3 Axis qualification vibration testing.
 - 4.3.13.4. Electronic Keys
 - 4.3.13.4.1. Electronic keys shall be designed to be programmed as user, control, setup, audit or operator plus audit options.
 - 4.3.13.4.2. Electronic keys shall have an operating temperature of -4 F to 122F (-20C - 50C)
 - 4.3.13.4.3. Electronic keys shall have rechargeable battery capable of 1,800 cylinder openings between charges.
 - 4.3.13.4.4. Electronic keys shall have a charging temperature range of 32F-104F (0C- 40C)
 - 4.3.13.4.5. Electronic Keys shall have battery level indication reporting a low battery at time of programming and during cylinder opening.
 - 4.3.13.4.6. Electronic Keys shall have hardened steel tip
 - 4.3.13.4.7. Electronic Keys shall be capable of storing up to 14,000 cylinders

- to be opened
- 4.3.13.4.8. Electronic Keys shall also store unauthorized attempts from Electronic Cylinders which shall be viewable in the software reporting once the key is updated/reconnected to the programming devices.
- 4.3.13.4.9. Electronic Keys shall use gold-plated, spring-loaded data contacts.
- 4.3.13.4.10. Electronic Keys audit reporting shall be capable of configuration in "first in first out" (FIFO) or to disable keys when memory is filled to prevent loss of audit trail data.
- 4.3.13.4.11. Electronic Keys shall store 14,000 audit events.
- 4.3.13.4.12. Electronic keys shall utilize hardened steel frame and plastic housing.
- 4.3.13.4.13. Provide (50) electronic keys with up to five different sets of programmed groups to be identified by VA by 95% Design Intent Drawing Review.
- 4.3.13.5. Wall Mounted Identification and Programming Device (IPD)
 - 4.3.13.5.1. IPD shall allow system administrators to manage the electronic key system anywhere within the VA campus or CBOC facilities with a VA network connection.
 - 4.3.13.5.2. Power Drop and Network drops shall be provided by contractor.
 - 4.3.13.5.3. IPD shall utilize LCD-based touchscreen for easy data input and system feedback, Linux-based OS for efficient operation, and hidden cable connection.
 - 4.3.13.5.4. IPD shall include optional four (4) to eight (8) digit personal identification number (PIN) for secondary authentication in high security environments.
 - 4.3.13.5.5. IPD shall provide programming feedback so that the user can visualize
 - 4.3.13.5.5.1. Key updates are complete
 - 4.3.13.5.5.2. Battery charge status
 - 4.3.13.5.5.3. Key errors
 - 4.3.13.5.5.4. Network connection status
 - 4.3.13.5.5.5. Provide (1) IPD at the main employee entrance or employee break area.
- 4.3.13.6. Software for the key system shall be compatible with the following and provided by the VA
 - 4.3.13.6.1. Software shall be self-hosted and TRM approved for use on the VA network.
 - 4.3.13.6.2. Software shall allow the programming of electronic cylinders and electronic keys either locally, or via remote programming devices.
 - 4.3.13.6.3. Software shall allow time stamp audit of electronic keys and electronic cylinders, either locally or via remote programming devices.
 - 4.3.13.6.4. Software shall allow the creation of individual users, or user groups to ensure ease of programming and maintenance of the electronic keys system.
 - 4.3.13.6.5. Software shall allow authorization of individual keys or user groups to specific time and date schedules, including the expiration of keys.
 - 4.3.13.6.6. Software shall allow the blacklisting of keys from electronic cylinders to eliminate risk association with lost keys and eliminate the need to rekey cylinders.
 - 4.3.13.6.7. Software to be hosted on servers provided to the VA based on "XT Self-Hosted System Requirements"
- 4.3.13.7. Door Hardware
 - 4.3.13.7.1. Doors shall have door handles or door pulls with heavyweight hinges. The Lessor is encouraged to avoid the use of chrome-plated hardware.
 - 4.3.13.7.2. All doors shall have corresponding doorstops (wall- or floor-mounted) and silencers.
 - 4.3.13.7.3. Automatic doors are also required on doors leading from the waiting room to the clinic areas.
 - 4.3.13.7.4. Exterior doors with removable hinge pins will be retained with set pins or spot

- welded.
 - 4.3.13.7.5. Hinge pins and hasps shall be secured against unauthorized removal by using spot welds or pinned mounting bolts.
 - 4.3.13.7.6. The exterior side of the door shall have a lock guard or astragal to prevent tampering of the latch hardware.
 - 4.3.13.7.7. Doors used for egress only shall not have any operable exterior hardware. All security- locking arrangements on doors used for egress shall comply with requirements of NFPA 101 or the International Building Code current as of the Lease Award Date. Egress doors must be equipped with panic hardware and meet NFPA requirements for exit.
 - 4.3.13.7.8. All patient exam rooms shall have privacy locks. See also. DA-149, VA Design Alert for Patient Privacy and Women Veterans' Health at <https://vatilms.va.gov/vatilms/reports/Alerts/Design%20Alerts>.
 - 4.3.13.7.9. All single-use toilet rooms shall have a privacy lock with occupancy indicator.
 - 4.3.13.8. Low Energy Automatic operators, ANSI A156.19-07. Heavy duty commercial grade. Provide complete with drop plates, bracket, or adapters for arms as required to suit details. Provide a terminal strip in an enclosed box near or above door that indicates connections for Security and Fire Alarm equipment and for electrified hardware items associated with proper door operation, as indicated by hardware group operational description. Coordinate with Security Contractor for doors actuated by electronic access control system.
 - 4.3.13.8.1. Public and Patient Toilets shall have automatic opening doors
 - 4.3.13.8.2. Doors separating the public waiting area to the staff or patient-care area shall have a vision panel and automatic opening device.
 - 4.3.13.9. Personal Access Control System (PACS): Electronic access control system/device(s), power supplies (unless otherwise noted in hardware group) and monitoring/alarm(s) are provided with Security System. General Contractor to coordinate the provision and installation of the products. Connection by Electrical.
 - 4.3.13.9.1. Location for PACS with Personal Identification Verification (PIV) card: Doors separating the public waiting area to the staff or patient-care areas (free exit to waiting or public area); Exterior doors.
 - 4.3.13.9.2. Location for PACS with two-factor authentication (2FA) (PIV and pin pad): Demarc Room, Telecommunication Room
 - 4.3.14. Patient Flag System. Add a 4-color mechanical arm flag indicator for all patient exam and procedure rooms.
 - 4.3.15. Toilet Rooms: All toilet rooms shall meet handicap accessibility codes and will include grab bars at commode. Metal partitions required at urinals and metal partitions with doors at commodes unless otherwise noted. All stalls shall be mounted from the ceiling/wall and not floor for infection prevention/cleaning purposes. Toilet rooms that include Baby Changing Stations must include the appropriate identification on the signage. Please refer to Design for Patient Privacy and Women Veterans' Health. All single stall restrooms, patient and/or staff, shall be noted as unisex.
- 4.4. Interior Design
- 4.4.1. General Criteria. Refer to Appendix: Room Data Matrix – Interior Construction for finish requirements listed by room type.
 - 4.4.2. Acoustic Ceiling Tile. AT: Acoustical ceiling tile, standard size to be 24"x24"x 1".
 - 4.4.3. Concrete Floor Treatment. Concrete Sealer (CS): 30 percent clear, non-yellowing, waterborne, membrane- forming curing and sealing compound: ASTM C1315, Type 1, Class A, minimum 30 percent total solids. Comply with ACI 301.
 - 4.4.4. Fiberglass-Reinforced Panels. Fiberglass-reinforced panels (FRP): Gelcoat-finished, glass-fiber reinforced plastic panels complying with ASTM D 5319. Fire-Rating per ASTM E 84: Class A. Nominal thickness: 0.09 inch.

4.4.5. Painting

- 4.4.5.1. Interior painting: Paint (P) and Epoxy Paint (EP): Paint surfaces with primer and two finish coats that are compatible with one another as demonstrated by manufacturer or field experience, unless otherwise indicated.
- 4.4.5.2. Materials: Use low VOC, low odor latex top coats where indicated. Paint the following items and surfaces:
 - 4.4.5.2.1. Ceilings: Flat latex.
 - 4.4.5.2.2. Walls - gypsum wallboard: Satin latex.
 - 4.4.5.2.3. Walls - concrete block: Semi-gloss latex with block filler.
 - 4.4.5.2.4. Hollow metal doors and frames: Semi-gloss latex over alkyd primer.
- 4.4.5.3. VA to select colors.

4.4.6. Wall Protection

- 4.4.6.1. Rigid Wall Covering (RWC): Nominal thickness .060 inch, high-impact, PVC-free, standard texture, chemical and stain resistant, with vertical, inside and outside trims. To comply with ASTM E84, ASTM F476 and ASTM D543.
- 4.4.6.2. Corner Guard (CG): Surface mounted assembly for angled corner consisting of a continuous aluminum retainer and vinyl/alloy extrusion in standard texture with ¼ inch corner radius. Size to be 2" width each way from corner. Height to be 7' from top of base.
 - 4.4.6.2.1. Location: Provide at all outside exposed corners in all clinic corridors outside of public areas as defined by FGI.
- 4.4.6.3. Corner Guard – Stainless Steel (CG-SS): Surface mounted stainless steel, 3-1/2" wing size, 7' height, 16 gauge in Kitchen and/or Standard Grade, #4 satin finish.
 - 4.4.6.3.1. Location: Provide at all outside exposed corners in loading dock area and heavy use equipment storage areas.

4.4.7. Window Shades

- 4.4.7.1. Manually operated window shade, chain driven, vertical roll-up, stainless steel bead chain with hold down clips, extruded aluminum rollers, and bottom slats. Aluminum headbox with endcaps and fascia with powder coat finish. Light gap reduction channels available for black out shades. PVC-free shades, Flame-Resistance Ratings: NFPA 701. Comply with WCMMAA100.1. Window treatments should not compromise patient safety. No draperies shall be installed.
 - 4.4.7.1.1. Locations:
 - 4.4.7.1.1.1. Semi-Transparent to be typical, unless otherwise noted.
 - 4.4.7.1.1.2. Black-out at conference rooms, tele-medicine rooms, optometry exam rooms.

4.4.8. Privacy Curtains / Screens

- 4.4.8.1. Provide curtain tracks with carriers and hooks (curtains shall be provided by the VA). Tracks shall be of extruded aluminum, ASTM B221, alloy 6063, temper T5 or T6, channel shaped, with smooth inside raceway for curtain carriers. End stop connectors, ceiling flanges and other accessories shall be fabricated from the same material with the same finish as the tracks or from nylon. Curtain tracks to be compatible with VA existing disposable curtains or Lessor provides curtain cleaning/replacement services. Confirm VA curtain compatibility during Design Phase.
- 4.4.8.2. Provide privacy curtains/screens to encompass adequate space for the healthcare provider to perform examination unencumbered by the curtain and provide a visually private patient changing area that allows the provider to remain in the room.
- 4.4.8.3. Refer to *Design for Patient Privacy and Women Veterans' Health* for proper locating in patient areas at <https://vatilms.va.gov/vatilms/reports/Alerts/Design%20Alerts>.
- 4.4.8.4. Locations shall be for all examination and procedure rooms.

4.5. Mechanical

4.5.1. References. The following publications shall be referenced for applicable systems calculations and design information: ASHRAE Handbooks.

4.5.2. HVAC Design Basis

4.5.2.1. Indoor design conditions:

4.5.2.1.1. Health care function design conditions shall be in compliance with the requirements of the FGI Guidelines and ASHRAE Standard 170, with the Room Data Matrix taking precedence and/or indicating required space design conditions when the FGI Guidelines do not fully address a space or room type.

4.5.2.1.2. Imaging Equipment Space Design Conditions

4.5.2.1.2.1. Equipment manufacturer requirements shall be followed.

4.5.2.1.2.2. Provide a dedicated computer room type AC unit to cool and control humidity for separate equipment rooms for radiology equipment.

4.5.2.1.2.3. Provide a computer room type AC unit to cool UPS rooms when segregated from other spaces.

4.5.2.1.3. Basic MEP Equipment Room Design Conditions:

- i. 4.7.2.1.5.1 Design conditions shall be as required to ensure tenant space design conditions are maintained.

4.5.2.1.4. Loading Docks:

4.5.2.1.4.1. Space shall have provisions to limit the intrusion airborne particulate and insects through dock door openings.

4.5.2.2. Outdoor design conditions:

4.5.2.2.1. Design with ASHRAE 2025 conditions associated with the ST LOUIS REGIONAL AP, IL, USA (WMO: 724395) location at: <https://ashrae-meteo.info/v3.0/>.

4.5.3. Air Handling Units (AHUs)

4.5.3.1. Air handling units shall consist of, but not be limited to, the following:

4.5.3.1.1. The units shall be constructed of double wall, insulated metal panels with an overall U-value and necessary features to meet/exceed energy code (and prevent condensation based on the site-specific outdoor design conditions).

4.5.3.1.2. Incorporate supply, return and outdoor air flow measuring stations for proper control feedback and ensure minimum ventilation needs.

4.5.3.1.3. For AHU's that are in excess of 15,000 cfm, SA/RA Array fans (minimum 2 fans per system) with backdraft isolation dampers shall be employed.

4.5.3.1.4. VSD controllers for fan array systems shall be provided in quantities that will provide no less than a 50% loss of nominal fan capacity if a single VSD were to fail.

4.5.3.2. For overall building capacity needs greater than 50,000 cfm, multiple AHUs shall be provided.

4.5.4. Exhaust Systems. Provide exhaust fans to provide general exhaust for toilet rooms, janitor's closets, soiled utility rooms and similar spaces.

4.5.5. Cooling System

4.5.5.1. Chillers shall employ multiple, independent refrigerant circuits, and shall have capacity control provisions to achieve at least 4 to 1 turndown.

4.5.5.2. The pumping system for the chilled water supply shall consist of variable speed pumps providing variable primary flow, arranged in a headered approach. Variable speed pumps shall each have a variable speed drive controller.

4.5.5.3. DX systems:

4.5.5.3.1. AHUs packaged with DX cooling sections shall have compressors/condensing units with direct drive scroll or screw

compressors, all-aluminum microchannel condenser coils, and direct drive TEFC condenser fans. Units shall employ multiple, independent refrigerant circuits, and shall have capacity control provisions to achieve at least 4 to 1 turndown.

- 4.5.5.3.2. Provide spring isolated mounting curbs for vibration attenuation, if applicable.
- 4.5.5.4. Provide unit supported screen wall panels, if roof supported structural screen wall elements are not incorporated.
- 4.5.5.5. Variable Refrigerant Flow (VRF) systems shall not be considered for VA CBOC projects due to various concerns such as inappropriate system type for healthcare facility applications due to higher amounts of fresh air requirement as well as safety risk to building occupants due to use of high refrigerant volumes and associated potential leaks.
- 4.5.6. Hot Water Heating System
 - 4.5.6.1. Hot water heating shall be generated by direct vent, sealed combustion, condensing type boilers.
 - 4.5.6.2. The heating water system distribution shall be a variable/primary pumping system. The pumping system will consist of variable speed pumps providing variable primary flow. Variable speed pumps shall each have a variable speed drive controller.
- 4.5.7. Humidification System
 - 4.5.7.1. Centralized humidification shall be provided when necessary to maintain 20%-60% relative humidity in all seasons. Each air handling unit shall have a dedicated humidifier.
 - 4.5.7.2. Water treatment shall be provided for humidifier feedwater as required by the humidifier equipment manufacturer.
- 4.5.8. HVAC Piping Systems
 - 4.5.8.1. Provide isolation valves on branch and lateral lines to reduce the amount of space that can be disrupted for maintenance and repair efforts. Provide isolation valves at each equipment connection.
 - 4.5.8.2. For hydronic systems utilized in specific sites with freezing climate considerations, use of propylene glycol and/or proven freeze protection methodologies must be incorporated.
 - 4.5.8.3. When propylene glycol is used, the freezing point of the glycol solution shall be at least 5°F lower than the minimum annual extreme daily temperature, to prevent the formation of crystals.
- 4.5.9. Air Distribution and Duct Systems
 - 4.5.9.1. Duct construction shall be as follows:
 - 4.5.9.1.1. All supply return and exhaust air ductwork shall be constructed of G90 galvanized steel. Non-ducted Return Air Plenum is not allowed. Flexible duct is prohibited except for the last three feet to connect diffusers.
 - 4.5.9.1.2. Ductwork shall be rectangular, round or flat oval constructed to SMACNA standards.
 - 4.5.9.1.3. Ductwork shall be constructed and tested with leakage classifications and pressure ratings and based on at least 125% of the actual operating pressure of the duct system.
 - 4.5.9.2. Grilles, registers and diffusers shall be steel, aluminum or extruded aluminum with appropriate finish.
- 4.5.10. HVAC Insulation Systems
 - 4.5.10.1. All exposed (visible in space) supply and return ductwork in the occupied conditioned spaces shall be provided with rigid insulation with proper seals. Painting and finish requirements shall be coordinated with the finish schedule.
- 4.5.11. Testing, Adjusting and Balancing
 - 4.5.11.1. An independent third-party NEBB, ABBC or TABB certified test and balance contractor shall be hired by the general contractor to balance and

document all air and hydronic systems within project scope. All ductwork shall be constructed and properly sealed in accordance with applicable energy code requirements. All ducts operating at 2 inches water gauge (wg) or greater shall be pressure tested based on code requirements for 3 in wg or greater pressure classification.

- 4.5.11.2. The balance contractor shall review all plans, components, access, etc. to ensure balancing activities may be successfully performed.
- 4.5.12. Automatic Temperature Control Systems
 - 4.5.12.1. The temperature controls shall be direct digital control (DDC) system with industry open protocol compatibility. The system shall provide control of the environment and other parameters and collect data on the performance and deviations (alarms).
 - 4.5.12.2. Dedicated thermal control zoning (thermostats) shall be provided for the following:
 - 4.5.12.2.1. each corner space,
 - 4.5.12.2.2. each conference room (or similar),
 - 4.5.12.2.3. each Imaging room,
 - 4.5.12.2.4. each Procedure room (or similar dedicated spaces),
 - 4.5.12.2.5. each logistics storage area
- 4.5.13. For typical blocks of rooms arranged together, every 3 exterior offices or exam rooms, or every 4 interior offices or exam rooms shall be permitted on a single zone.
- 4.5.14. Special Conditions
 - 4.5.14.1. Room Pressurization: Health care function design conditions shall be in compliance with the requirements of ASHRAE 170, with the Room Data Matrix taking precedence and/or indicating required space design conditions when ASHRAE 170 does not fully address a space or room type.
 - 4.5.14.1.1. Positive air balance is designated as (+)
 - 4.5.14.1.2. More positive air balance is designated as (++)
 - 4.5.14.1.3. Negative air balance is designated as (-)
 - 4.5.14.1.4. More negative air balance is designated as (--)
 - 4.5.14.1.5. If the room is not indicated with a specific pressure relationship (blank) or is indicated with *NA* then there is no requirement.
 - 4.5.14.2. Imaging Areas:
 - 4.5.14.2.1. For HVAC ducts, pipes and devices penetrating shielded walls and ceilings, ensure coordination with the architectural discipline and provide treatment as specified by the equipment manufacturer and medical physicist.
 - 4.5.14.2.2. Provide booster humidification systems as may be necessary to comply with acceptable environmental ranges as prescribed by equipment manufacturers.
 - 4.5.14.3. Temperature, Humidity and/or Air-Balance sensitive areas shall not be on a set-back (unoccupied) program, and shall maintain air metric criteria listed on the Room Data Matrix 24/7.
 - 4.5.14.3.1. Real time monitoring of these conditions shall be provided through the HVAC direct digital control system, providing alarms via the control system when rooms are out of the specified acceptable range, triggering a lessor response.
 - 4.5.14.3.2. Locations: Medication Room, Clean Utility Room, and Telecommunication Room.

4.6. Plumbing

- 4.6.1. References. The following publications shall be referenced for applicable systems calculations and design information: ASPE Handbooks.
- 4.6.2. Plumbing Design Basis
 - 4.6.2.1. Domestic Water Service

- 4.6.2.1.1. Incoming service water pressure shall be ascertained to determine if adequate pressure is available on site to serve the facility without the use of booster pumps. Coordinate with the water supplier to provide and evaluate historical water data during all seasons.
- 4.6.2.1.2. If booster pumps are required, they shall utilize variable speed drives for pressure control.
- 4.6.2.2. Domestic Water Heating
 - 4.6.2.2.1. If storage type heaters are utilized, they shall be set to 140 degrees F and a thermostatic mixing valve incorporated to temper the water to 115 degrees for domestic use.
 - 4.6.2.2.2. Provide point of use mixing valves to ensure water temperature is no greater than 110F.
 - 4.6.2.2.3. A looped hot water recirculation system with a bronze in-line recirculation pump shall be provided.
- 4.6.2.3. Domestic Water Treatment
 - 4.6.2.3.1. The facility water system shall have a building program for risk assessment and Legionella mitigation provisions as needed to comply with ASHRAE Standard 188.
 - 4.6.2.3.2. Water systems supplying critical water for SPS, or water for Dialysis shall meet the requirements of AAMI.
- 4.6.2.4. Sump Pumps and Sewage Ejectors
 - 4.6.2.4.1. If required, 50/50 duplex submersible sump pump and/or sewage ejector systems shall be utilized.
- 4.6.2.5. Fire Suppression. Install a fire suppression system in accordance with NFPA 13, latest edition.
- 4.6.3. Materials
 - 4.6.3.1. Domestic Water Piping:
 - 4.6.3.1.1. Provide isolation valves on branch and lateral lines to reduce the amount of space that can get disrupted for maintenance and repair efforts. Also provide isolation valves at each equipment connection.
 - 4.6.3.2. Sanitary, Storm, Vent, and Clear Water Waste Systems
 - 4.6.3.2.1. Aboveground:
 - 4.6.3.2.1.1. Cast-Iron pipe, with compatible joints and fittings.
 - 4.6.3.2.1.2. Type DWV Copper Tube, with compatible joints and fittings.
 - 4.6.3.2.1.3. Type 304 or 316 stainless steel pipe, with compatible joints and fittings.
 - 4.6.3.2.1.4. Schedule 40 PVC pipe, solid wall, with compatible joints and fittings.
 - 4.6.3.2.2. Underground:
 - 4.6.3.2.2.1. Cast-Iron pipe, with compatible joints and fittings.
 - 4.6.3.2.2.2. Type DWV Copper Tube, with compatible joints and fittings.
 - 4.6.3.2.2.3. Type 304 or 316 stainless steel pipe, with compatible joints and fittings.
 - 4.6.3.2.2.4. Schedule 40 PVC pipe, solid wall, with compatible joints and fittings.
 - 4.6.3.3. Provide overflow roof drain inlets and piping system independent to storm water piping serving roof drains.
 - 4.6.3.4. Floor drains shall not be provided in single user toilet rooms unless required by local or state code provisions.
 - 4.6.3.5. Plumbing Fixtures
 - 4.6.3.5.1. Fixtures shall be commercial or institutional grade.
 - 4.6.3.5.2. An accessible public refrigerated, wall hung drinking fountain will be provided in the waiting room. Drinking fountain model shall contain a bottle filter feature.
 - 4.6.3.6. Hot Water Supply and Recirculation Piping:
 - 4.6.3.6.1. Insulation systems shall conform to applicable Energy Code

requirements and current NAIMI standards.

4.6.3.6.2. Insulation on hot piping shall have thickness sufficient to prevent skin injury due to excessive temperature.

4.6.3.7. Cold Water Supply Piping: Insulation shall include vapor retarders and shall have thickness sufficient to prevent condensation.

4.6.3.8. Rainwater Piping: Insulation shall include vapor retarders and shall have sufficient thickness to prevent condensation.

4.6.3.9. Outdoor exposed insulation shall be protected from the elements with a fully sealed outer covering that is UV protected and has a minimum puncture resistance rating of 55 lbs.

4.6.4. Medical Gases. There are no piped medical gas requirements.

4.6.5. Special Conditions

4.6.5.1. Imaging Areas: For pipes and devices penetrating shielded walls and ceilings, ensure coordination with the architectural discipline and provide treatment as specified by the equipment manufacturer and medical physicist.

4.7. Electrical

4.7.1. References. All electrical systems shall meet and be in compliance with the National Electric Code (NFPA 70).

4.7.2. Generator. There are no emergency or standby generator requirements,

4.7.3. Loads shall be determined as per the requirements of the room data matrix.

4.7.4. Lighting

4.7.4.1. General: At a minimum, lighting systems design and installation shall be designed based on latest edition of the IESNA Handbook, as well as applicable IESNA standards.

4.7.4.2. Indoor Lighting Specifics. For specific room lighting requirements above and beyond IESNA's guidance, refer to Room Data Matrix.

4.7.5. Lighting Controls:

4.7.5.1. Local dual level switching or dimming will be provided allow occupant selection of lighting level. Locations: Exam Rooms, Tele-health/retinal Rooms, Consult Rooms, Offices, Group Room, Conference Room.

4.7.5.2. Exterior lighting will be controlled by exterior photocell and astronomic timeclock input through the lighting control relay panel system.

4.7.6. LED Drivers

4.7.6.1. Minimum efficiency of 85%

4.7.6.2. 20% THD or less

4.7.7. Special Requirements

4.7.7.1. Provide HDMI wall port for workstation connection to the TV monitors in the Conference Room and Group Room/Whole Health.

4.7.7.2. Corridors will have outlets at least every 15 feet.

4.7.7.3. All outlets in all rooms to be set ground plug up.

4.7.7.4. All patient-care rooms shall have hospital-grade receptacles.

4.7.7.5. Lessor shall provide a 120V power and a data outlet for the message board. Outlet shall be located 6" below the finished ceiling at a waiting room location identified during the design phase.

4.8. Telecommunications

4.8.1. Scope of Work

4.8.1.1. The Lessor shall provide the following:

4.8.1.1.1. Telephone cabling, pathways (conduit and cable tray), outlets, faceplates, terminal blocks, backboards, cable terminations and cable testing.

4.8.1.1.2. Data cabling (fiber optic and copper), pathways (conduit and cable tray), outlets, faceplates, patch panels, network equipment racks, network equipment cabinets, cable terminations and cable testing.

- 4.8.1.2. The VA IT department will provide the following:
 - 4.8.1.2.1. Telephone System hardware and electronics such as voice mail servers and telephone handsets.
 - 4.8.1.2.2. Data network electronics such as concentrators, Ethernet switches, servers, PCs, Wireless Access Points and other electronic equipment.
- 4.8.2. Pathways
 - 4.8.2.1. Boxes and Conduits
 - 4.8.2.1.1. Voice and data outlets shall be provided with a 4" square by 2-1/8" deep box with a single gang, telecommunication rated work box with a minimum of a 3/4" conduit routed up to an accessible ceiling space.
 - 4.8.2.2. Sleeves
 - 4.8.2.2.1. Where cables penetrate through walls, conduit sleeves with bushings on both ends, shall be provided. All penetrations through fire rated walls shall be fire stopped.
 - 4.8.2.2.2. Where cables penetrate through floors of telecommunications rooms, a minimum of four (4) 4-inch conduit sleeves with bushings on both ends, shall be provided. All penetrations through floors shall be fire stopped.
 - 4.8.2.2.3. Conduit sleeves shall be sized to be filled with cables to no more than 40 percent of the cross-sectional area of the conduit.
 - 4.8.2.3. Cable Support
 - 4.8.2.3.1. Wire mesh cable tray a minimum of 4" deep x 12" wide shall be provided and supported from the structural steel or concrete structure with a minimum of 3/8" diameter threaded rods to support the horizontal and backbone communications cables along the main pathways above the suspended ceiling space. In finished spaces without a suspended ceiling, provide a minimum of 4" deep x 12" wide Solid Bottom cable tray instead of Wire Mesh cable tray along the main pathways. Cable trays shall be sized to be filled with cables to no more than 50% of the cross-sectional area of the cable tray. Where the cable tray fill ratio exceeds 50% of the cable tray cross-sectional area, provide a larger cable tray or two cable trays.
 - 4.8.2.3.2. A minimum of 12 inches of free access shall be provided and maintained above the cable trays and along one side of the cable trays.
 - 4.8.2.3.3. Where cables are routed in the open outside of the cable tray above the suspended ceiling space, adequate cable support via J-hooks shall be located at a maximum of 48" intervals.
 - 4.8.2.3.4. As per National Electrical Code and TIA-569 standard, the suspended ceiling support wires or support rods shall not be used as a means of cable support. Cables shall not be laid directly on the ceiling tile, ceiling grid rails, or on the structural steel (bar joists). An independent hanger system shall be used.
- 4.8.3. Spaces
 - 4.8.3.1. Entrance Facility Room (AKA Demarcation Point- Demarc)
 - 4.8.3.1.1. The Entrance Facility Room is the location where the Local Exchange Carriers and other communications Service Providers such as telephone, data, and MATV/CATV install their cabling and equipment to bring services into the building. It also establishes the physical point where the service provider's responsibilities for service and maintenance end.
 - 4.8.3.1.2. The minimum size of the Entrance Facility Room shall be 80 sq. ft. with an additional 20 sq. ft. for every additional rack required.
 - 4.8.3.1.3. The room shall not be located directly below or adjacent to laboratories, kitchens, laundries, rest rooms, showers, or other facilities where water service is provided.
 - 4.8.3.1.4. Any pipe or duct system foreign to the room installation shall not

enter or pass through the room. The design professional shall ensure that foreign piping such as water pipes, steam pipes, medical gas pipes, sanitary waste pipes, roof drains, AC ducts, and other unrelated piping containing liquids or gases are not installed or pass through the room. Sprinkler piping shall not be routed through the room, unless it serves to protect the installation.

- 4.8.3.1.5. A minimum of (2) 4" conduits shall be installed out to the property line to provide pathways for the services providers to install their cabling.
- 4.8.3.1.6. All the walls of the room shall be constructed from drywall deck to deck, not just from floor to suspended ceiling height. All the walls of the room shall be covered from the floor to a minimum height of 8'-0" above the floor with 3/4-inch exterior AC grade flame retardant plywood and painted a light color to reflect the room light and reduce dust.
- 4.8.3.1.7. The lighting shall be a minimum of 500 lux in the horizontal plane and 200 lux in the vertical plane when measured at 3 feet above the finished floor.
- 4.8.3.1.8. The door shall be a minimum of 36 inches wide and 96 inches high, hinged to open outward and fitted with a card reader security lock.
- 4.8.3.1.9. Protective cages shall be installed on all water-based fire protection sprinkler heads located within the room.
- 4.8.3.1.10. The Service Provider IT equipment installed in the Entrance Facility Room will be required to operate 24 hours a day and 365 days a year. The HVAC system shall be designed and installed to maintain a room temperature of 64-75 degrees Fahrenheit and relative humidity of 30-55 percent noncondensing on a 24-hour basis.
- 4.8.3.2. Telecommunication Room (TR)
 - 4.8.3.2.1. The TR is a centralized space for telecommunications and computer equipment that serves an entire building. Typical equipment includes phone switches, voice mail servers, file/application servers, video surveillance storage and Core Ethernet switches. There shall be at least one Telecommunications Room on each floor. Each work area shall be served by a Telecommunications Room that is located on the same floor that the work area is located. There shall be a minimum of one (1) Telecommunications Room in each building.
 - 4.8.3.2.2. The location of the TR should be determined after careful consideration. Locations should be avoided that restrict expansion of the room due to building construction such as elevators, mechanical rooms, core hallways, outside walls, or other fixed building walls. The location should consider accessibility requirements for the delivery of large equipment to the room and be located away from EMI sources that limit EMI field strength to no more than 3.0 V/m throughout the frequency spectrum.
 - 4.8.3.2.3. The location of the Telecommunications Rooms shall be as close as possible to the central core of the building floor to keep horizontal cable lengths to a minimum. Additional Telecommunications Rooms shall be provided where the horizontal cable length from the telecommunications room to the farthest workstation location exceeds 90 meters (295 feet).
 - 4.8.3.2.4. Telecommunications Rooms located on the same floor shall be no farther than 150 meters (500 feet) apart to limit horizontal cable lengths to 90 meters (295 feet) or less. Provide secondary TR if needed geographically.
 - 4.8.3.2.5. The room shall not be located directly below or adjacent to laboratories, kitchens, laundries, rest rooms, showers, or other facilities where water service is provided.
 - 4.8.3.2.6. Any pipe or duct system foreign to the room installation shall not

enter or pass through the room. The design professional shall ensure that foreign piping such as water pipes, steam pipes, medical gas pipes, sanitary waste pipes, roof drains, AC ducts, and other unrelated piping containing liquids or gases are not installed or pass through the room. Sprinkler piping shall not be routed through the room, unless it serves to protect the TR installation.

- 4.8.3.2.7. The TR shall be dedicated to telecommunications and computer equipment. The room shall not be shared with electrical equipment, heating/ventilating and air conditioning equipment, fire detection systems, or other mechanical systems unless these systems are specifically needed and dedicated to support the computer room and its functions.
- 4.8.3.2.8. The minimum sizes of the TR shall be 170 SF. The size should be confirmed with specific IT equipment sizes. The size also assume there are no obstructions in the room such as columns and the rooms are a rectangular shape.
- 4.8.3.2.9. All the walls of the TR shall be constructed of drywall deck to deck, not just from the floor to the suspended ceiling height. The floor, walls, and ceiling shall be sealed/painted to reduce dust and shall be light colored to reflect room light.
- 4.8.3.2.10. Flooring materials shall be used that have antistatic properties. The floor shall be covered with light colored luxury vinyl floor tile to reflect the room light and reduce dust.
- 4.8.3.2.11. At least one of the walls of the room shall be covered from the floor to a minimum height of 8'-0" above the floor with 3/4-inch exterior AC grade flame retardant plywood and painted a light color to reflect the room light and reduce dust.
- 4.8.3.2.12. 18" wide ladder racking cable tray shall be provided and supported from the structure above with 3/8" diameter threaded rods over the equipment racks and the side walls to support the horizontal and backbone communications cables through-out the Telecommunications Room.
- 4.8.3.2.13. The lighting shall be a minimum of 500 lux in the horizontal plane and 200 lux in the vertical plane when measured at 3 feet above the finished floor in between all rows of equipment cabinets and equipment racks.
- 4.8.3.2.14. The door shall be 36 inches wide and 96 inches high and fitted with a card reader security lock. Where the *Generic Extra Small Campus Support Center (Data Center) Design Portfolio* has been specified, this shall be a double door 72 inches wide and 96 inches high, without a central mullion.
- 4.8.3.2.15. Protective cages shall be installed on all water-based fire protection sprinkler heads located within the TR.
- 4.8.3.2.16. The IT equipment installed in the Telecommunications Room will be required to operate 24 hours a day and 365 days a year. The HVAC system shall be designed and installed to maintain a room temperature of 64-75 degrees Fahrenheit and relative humidity of 30-55 percent non-condensing on a 24-hour basis.

4.8.4. Telecommunications Bonding and Grounding

4.8.4.1. Telecommunications Primary Bonding Busbar

- 4.8.4.1.1. Each Entrance Facility Room shall contain a Telecommunications Primary Bonding busbar for providing a central location for bonding all telecommunications equipment in the Entrance Facility Room per the TIA-607-D Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises, local codes, the VA Electrical Design Manual, and National Electrical Safety Code.
- 4.8.4.1.2. The Telecommunications Primary Bonding busbar shall consist of a

predrilled copper busbar with TIA-607 standard sizing and spacing. It shall have minimum dimensions of ¼ inch thick, 4 inches wide, and the length shall be a minimum of 23 inches. The bonding busbar shall be insulated from its support by a minimum of a 2-inch separation.

- 4.8.4.1.3. Building structural steel (beams and/or columns) within 6 feet of the bonding busbar shall be bonded to the bonding busbar with a minimum of a 6 AWG copper conductor.
- 4.8.4.2. Telecommunications Secondary Bonding Busbar
 - 4.8.4.2.1. Each TR shall contain a Telecommunications Secondary Bonding Busbar for providing a central location for bonding all telecommunications equipment in the room per the TIA-607 standard.
 - 4.8.4.2.2. The Telecommunications Secondary Bonding Busbar shall consist of a predrilled copper busbar with TIA-607 standard sizing and spacing. It shall have minimum dimensions of ¼ inch thick, 2 inches wide, and the length shall be a minimum of 12 inches. The bonding busbar shall be insulated from its support by a minimum of a 2-inch separation.
 - 4.8.4.2.3. Building structural steel (beams and/or columns) within 6 feet of the bonding busbar shall be bonded to the bonding busbar with a minimum of a 6 AWG copper conductor.
- 4.8.4.3. Telecommunications Rack Bonding Busbar
 - 4.8.4.3.1. Racks located in the Entrance Facility Room, TR, as well as Telecommunications Enclosures (TEs) shall have a horizontal Rack Bonding Busbar installed in the top of the rack/cabinet in RU 45 (install in the top rear position in TEs) to provide effective bonding of the rack/TE to the Primary bonding busbar or Secondary Bonding Busbar and provide a central location for the bonding of all telecommunications equipment located in the rack/TE per the TIA-607 standard. The busbar shall consist of a pre-drilled copper busbar with TIA-607 standard sizing and spacing.
 - 4.8.4.3.2. The Rack Bonding Busbar shall be bonded to the Telecommunications Primary Bonding busbar or Telecommunications Secondary Bonding busbar in the room with a minimum of a 6 AWG copper conductor.
 - 4.8.4.3.3. Rack mounted IT equipment with integral bonding terminals shall be bonded to the Rack Bonding Conductor (RBC) or to a vertical/horizontal Rack Bonding Busbar (RBB). An RBC is a bonding conductor from the rack or RBB to the TEBC. Each cabinet or equipment rack will have a suitable connection point to which the bonding conductor can be terminated. Properly sized listed two-hole compression lugs or listed terminal blocks with two internal hex screw or equivalent torque characteristics shall be used at the connection point.
- 4.8.4.4. Telecommunications Bonding Backbone Cable
 - 4.8.4.4.1. The Telecommunications Primary Bonding Busbar in the Entrance Facility room and Telecommunications Secondary Bonding Busbars in the TR shall be bonded to the building grounding electrode system with a bonding backbone cable that is a minimum of a 3/0 AWG stranded copper conductor.
 - 4.8.4.4.2. The building structural steel shall not be used as a replacement for the bonding backbone cable.
- 4.8.4.5. Bonding of Cable Tray and Equipment
 - 4.8.4.5.1. Cable tray and/or Ladder racking shall be bonded to the Primary Bonding Busbar or Secondary Bonding Busbar with a minimum of an insulated #6 AWG stranded copper conductor and connectors designed for the specific purpose.
 - 4.8.4.5.2. Bonding of other telecommunications equipment in the TR to the bonding busbars shall be executed as required by the equipment manufacturer.

4.8.5. Equipment Racks and Equipment Cabinets

4.8.5.1. Ensure that IT equipment racks are installed flush to one another without air gaps between the racks. Use appropriate materials to fill gaps between the racks to prevent recirculation of exhaust air to the cold aisle. Meet the following requirements:

- 4.8.5.1.1. Style: Channel
- 4.8.5.1.2. Height: 84 inches, Width: 24 inches, Depth: 30 inches minimum.
- 4.8.5.1.3. Equipment Mounting Width: 19 inches.
- 4.8.5.1.4. Equipment Mounting Height: 45 RUs.
- 4.8.5.1.5. Front and Rear rails: EIA threaded or Square Holes for cage nuts.
- 4.8.5.1.6. Rail Marking: Rack unit markings present on front and rear rails starting at one RU at the bottom.
- 4.8.5.1.7. Weight Capacity: 2,500 lbs. minimum.
- 4.8.5.1.8. Cable Management: Built-in overhead water fall and cable management strap attachment points.
- 4.8.5.1.9. Seismic bracing where required by Code.
- 4.8.5.1.10. Provide rack PDU brackets.
- 4.8.5.1.11. Color: White.

4.8.5.2. Telecommunications Enclosures (TEs) installed shall meet the following requirements:

- 4.8.5.2.1. NEMA-12 or equivalent construction. Dust seals and replaceable inlet/outlet filters for vents/airflow openings/fans provided. This is required regardless of planned installation environment.
- 4.8.5.2.2. Filters shall be commercially widely available and initially provided with the TE.
- 4.8.5.2.3. Environmentally controlled enclosures are acceptable.
- 4.8.5.2.4. 24" minimum width to allow for power and telecommunications cabling management to the sides of rack-mounted equipment.
- 4.8.5.2.5. 24RU in height or larger.
- 4.8.5.2.6. Unit mounts to ¾" plywood backboard via 16" OC mounting for standard stud construction.
- 4.8.5.2.7. Unit opens in rear (swings open) for access to rear of installed equipment. Unit opens in front (swinging front door) for access to front of installed equipment. Both sections are able to be physically locked.
- 4.8.5.2.8. Adjustable 19" EIA/TIA rack rails. Rear rail kits are required.
- 4.8.5.2.9. Top and bottom knockouts for cable/conduit entry. All knockouts must be sealable and sealed for liquid and dust entry resistance. The use of a knockout kit to create larger penetrations is acceptable.
- 4.8.5.2.10. 115V fans to remove heat generated in TE are required. Whether these are used as exhaust, intake, or both is not specified.
- 4.8.5.2.11. Provide TEs with fiber distribution cabinets, fiber cassettes, UTP patch panels, horizontal cable management units, and shelves as required for the specific implementation.

4.8.6. Power Distribution Units (PDUs) and Uninterruptable Power Supplies (UPSs)

4.8.6.1. Equipment Rack and Equipment Cabinet 120/208 Volt PDUs for TRs

- 4.8.6.1.1. Input: 20 Amp Three-phase; 120/208V, L21-20P Plug.
- 4.8.6.1.2. Circuit Breakers: 3 x 2 Pole 20 Amp Hydraulic Magnetic breakers.
- 4.8.6.1.3. Receptacles: (30) C13 receptacles 208 Volt, (6) C19 receptacles 208 Volt, (2) 5-20 receptacles 120 Volt.
- 4.8.6.1.4. IP and Serial monitoring.
- 4.8.6.1.5. Ethernet, USB, and Environmental sensor ports.
- 4.8.6.1.6. Mounting: Vertically on the rear rails of the rack.
- 4.8.6.1.7. Quantity: Provide two (2) PDUs in each equipment rack in Telecommunications Rooms.

4.8.6.2. Equipment Rack and Equipment Cabinet 120/208 Volt PDUs for TRs

- 4.8.6.2.1. Input: 20 Amp Three-phase; 120/208V, L21-20P Plug.

**Madison County VA Clinic
Agency Specific Requirements**

- 4.8.6.2.2. Circuit Breakers: 3 x 2 Pole 20 Amp Hydraulic Magnetic breakers.
- 4.8.6.2.3. Receptacles: (30) C13 receptacles and (6) C19 receptacles, 208 Volt.
- 4.8.6.2.4. IP and Serial monitoring.
- 4.8.6.2.5. Ethernet, USB, and Environmental sensor ports.
- 4.8.6.2.6. Mounting: Vertically on the rear rails of the rack.
- 4.8.6.2.7. Quantity: Provide two (2) PDUs in each equipment rack and each equipment cabinet in TR.
- 4.8.6.3. Equipment Rack and Equipment/Cabinet UPSs for TRs
 - 4.8.6.3.1. Input: 20 Amp Three-Phase; L21-20P Plug.
 - 4.8.6.3.2. Output: One (1) L21-20R receptacle
 - 4.8.6.3.3. Capacity: 5 kW
 - 4.8.6.3.4. Run time at full capacity: Minimum of 10 minutes.
 - 4.8.6.3.5. Mounting: Rack or Cabinet 19-inch TIA-310 mounting width.
 - 4.8.6.3.6. Quantity: Provide one (1) UPS in each equipment rack and each equipment cabinet in TRs. (No rack-mounted UPSs are installed in the Extra Small Campus Support Center (Data Center) specified for VA-occupied spaces greater than 50,000 sq. ft.)
- 4.8.6.4. Equipment Rack and Equipment/Cabinet UPSs for TEs
 - 4.8.6.4.1. Input: 20 Amp single-phase L5-20P Plug.
 - 4.8.6.4.2. Output: One (1) L5-20R receptacle
 - 4.8.6.4.3. Capacity: 2 KW
 - 4.8.6.4.4. Run time at full capacity: Minimum of 10 minutes.
 - 4.8.6.4.5. Mounting: Rack or Cabinet 19-inch TIA-310 mounting width.
 - 4.8.6.4.6. Quantity: Provide one (1) UPS in each Telecommunications Enclosure (TE).
- 4.8.6.5. Zone Power Distribution Units (PDUs)
 - 4.8.6.5.1. TRs with more than one equipment rack and/or more than one equipment cabinet, shall be provided with Zone PDUs used for power distribution to the rack mounted and cabinet mounted PDUs and UPSs.
 - 4.8.6.5.2. Input: Two (2) 30 Amp Three-Phase L21-30P plugs. Power cords on the Zone PDU shall be of sufficient length to reach the supply branch circuit receptacles suspended over the rack or cabinet.
 - 4.8.6.5.3. Output: Four (4) L21-20R receptacles.
 - 4.8.6.5.4. Quantity: One (1) Zone PDU for every two (2) equipment racks. One (1) Zone PDU for every two (2) equipment cabinets.
 - 4.8.6.5.5. Supply Branch Circuits: Provide two (2) 30 amp 3-phase 120/208 Volt (Wye) circuits with L21-30R receptacles for each Zone PDU. If a Generator is installed at the site, connect the branch circuits to a Panelboard connected to the Generator. Suspend the receptacles over the equipment racks/cabinets from the ceiling for each Zone PDU.
- 4.8.6.6. Equipment Rack and Equipment Cabinet 120 Volt PDUs for TEs
 - 4.8.6.6.1. Input: 20 Amp 120 Volt, NEMA L5-20P Plug.
 - 4.8.6.6.2. Receptacles: Minimum of eight (8) 5-15/20R receptacles.
 - 4.8.6.6.3. Mounting: Horizontal in Rack or Cabinet 19-inch TIA-310 mounting width.
 - 4.8.6.6.4. Quantity: Provide 2 horizontal rackmount PDUs when a Telecommunications Enclosure (TE) is specified.
- 4.8.7. Telecommunications Infrastructure Plant (TIP)
 - 4.8.7.1. Horizontal Cabling
 - 4.8.7.1.1. Cable
 - 4.8.7.1.1.1. The horizontal cabling shall consist of a minimum of two (2) Category 6A UTP LP rated cables to each work area outlet for voice and/or data. The color of the cable jacket shall be blue.
 - 4.8.7.1.1.2. The horizontal cabling shall consist of a minimum of two (2) Category 6A UTP LP rated cables to each wireless LAN outlet. The

- color of the cable jacket shall be blue.
- 4.8.7.1.1.3. The length of the horizontal cables shall not exceed 90 meters (295 feet) from the telecommunications room to the work area outlet or the wireless LAN outlet.
- 4.8.7.1.1.4. Provide plenum rated cable above ceilings used as a return air plenum.
- 4.8.7.1.2. Workstation Outlets
 - 4.8.7.1.2.1. Each Category 6A horizontal cable shall be connected to category 6A RJ45 jacks at work area outlets.
 - 4.8.7.1.2.2. Each Category 6A horizontal cable shall be connected to category 6A RJ45 plugs at wireless LAN outlets.
 - 4.8.7.1.2.3. The pin configuration for each RJ45 jack shall conform to the TIA/EIA T568B standard.
 - 4.8.7.1.2.4. Refer to room matrix for work area outlet locations/quantities.
 - 4.8.7.1.2.5. One (1) wireless LAN outlet shall be provided for each 625 square feet of floor space.
 - 4.8.7.1.2.6. The typical standard density work area outlets will consist of two RJ45 interfaces. This provided connectivity for one IP telephone and one workstation.
- 4.8.7.1.3. Patch Cords
 - 4.8.7.1.3.1. Patch cords shall be factory terminated and shall match the category of the associated patch panel, work area outlet, and horizontal cable.
- 4.8.7.1.4. Cable Termination Hardware
 - 4.8.7.1.4.1. The Category 6A UTP horizontal cables shall be connected, in the Telecommunications Room, to Category 6A RJ45 48 port rack mounted angled patch panels. Angled patch panels containing more than 48 ports shall not be used. The pin configuration for each RJ45 jack shall conform to the TIA/EIA T568B standard.
 - 4.8.7.1.4.2. The horizontal cables shall be continuous from the angled patch panels to the work area outlet jacks and wireless LAN outlet plugs.
 - 4.8.7.1.4.3. The 48 port angled patch panels shall be mounted in 19-inch floor mounted equipment racks that are 84 inches tall. Wall mounted racks shall not be used except in facilities under 3,000 sq. ft.
 - 4.8.7.1.4.4. Front and rear six (6) inch wide vertical cable managers shall be installed on each side of the 19-inch equipment racks on the end of the row of racks and ten (10) inch wide vertical cable managers shall be installed between each rack.
 - 4.8.7.1.4.5. No more than eight (8) 48 port angled patch panels shall be installed in a single 84-inch-tall equipment rack. This allows for the lower half of the equipment rack to be used to mount Ethernet switches, UPS equipment and other network electronics. If more than eight (8) 48 port angled patch panels are required to terminate the horizontal cabling, then another equipment rack shall be installed.
- 4.8.7.2. Backbone Cabling
 - 4.8.7.2.1. Cable
 - 4.8.7.2.2. The backbone cable between TR shall consist of a minimum of one 25-pair Category 5e UTP copper cable and 24 strands of 850 nm laser-optimized (OM4) 50/125 multimode fiber optic cable.
 - 4.8.7.2.3. The backbone cable from the Entrance Facility Room to the TR shall consist of a minimum of 100 pairs of Category 5e UTP copper cable and 24 strands of 850 nm laser optimized (OM4) 50/125 multimode fiber optic cabling and terminations.
 - 4.8.7.2.4. Provide plenum rated cable above ceilings used as a return air plenum.

- 4.8.7.2.5. Provide indoor armored fiber optic backbone cable or provide unarmored fiber optic backbone cable installed in inner duct.
- 4.8.7.2.6. Copper Backbone Cable Termination Hardware
 - 4.8.7.2.6.1. The Category 5e UTP copper backbone cable shall be connected, in the Telecommunications Rooms, to 24-port rack mounted angled patch panels. (48-port angled patch panels are acceptable if more than one 25-pair backbone cable is specified.) Patch panels containing more than 48 ports shall not be used. One pairs of the backbone cable shall be terminated on each patch panel port (two pairs on port 24).
 - 4.8.7.2.6.2. The RJ45 angled patch panels shall be mounted in 19-inch floor mounted equipment channel racks that are 84 inches tall. Wall mounted racks (TEs) shall not be used except in facilities under 3,000 sq. ft.
 - 4.8.7.2.6.3. Front and rear six (6) inch wide vertical cable managers shall be installed on each side of the 19-inch equipment racks on the end of the row of racks and ten (10) inch wide vertical cable managers shall be installed between each rack.
 - 4.8.7.2.6.4. No more than eight (8) 48 port angled patch panels shall be installed in a single 84-inch-tall equipment rack. This allows for the lower half of the equipment rack to be used to mount Ethernet switches, UPS equipment and other network electronics. If more than eight (8) 48 port angled patch panels are required to terminate the backbone cabling, then another equipment rack shall be installed.
- 4.8.7.2.7. Fiber Optic Backbone Cable Termination Hardware
 - 4.8.7.2.7.1. The multimode fiber optic backbone cables shall be connected at each end to fiber optic cable connectors in one rack position height angled high density fiber distribution panels located in the TR, or Entrance Facility room. The high-density patch panels shall have the capacity to terminate a minimum of 144 strands of fiber optic cabling.
 - 4.8.7.2.7.2. All fiber optic backbone cable strands shall be terminated on fiber optic connectors. No fiber strands shall be left unterminated.
 - 4.8.7.2.7.3. All multimode fiber optic cables shall be factory pre-terminated with Multi-Fiber Push On (MPO) connectors in the Method A (straight through) polarity configuration. The MPO connector at each end of the cable shall be connected to a fiber optic cassette to provide duplex-LC connectors at both ends that will support one or two 12-strand cable assemblies or one 24-strand cable assembly for a total of 6 or 12 duplex-LC connectors. Cassettes can also be used, if directed by the VA IT department, to breakout a 24-strand backbone trunk into 3 MPO connectors of 8 fiber strands each to support 40 Gigabit Ethernet parallel transmission capabilities.
 - 4.8.7.2.7.4. The fiber distribution panels shall be mounted in 19-inch floor mounted equipment racks that are 84 inches tall. Wall mounted racks and/or wall mounted fiber distribution panels shall not be used.
 - 4.8.7.2.7.5. Front and rear six (6) inch wide vertical cable managers shall be installed on each side of the 19-inch equipment racks on the end of the row of racks and ten (10) inch wide vertical cable managers shall be installed between each rack.
 - 4.8.7.2.7.6. No more than twelve (12) one rack position height angled high density fiber distribution panels shall be installed in a single 84-inch-tall equipment rack. This allows for the lower half of the equipment rack to be used to mount Ethernet switches, UPS equipment and other network electronics. If more than twelve (12) one rack position height fiber distribution panels are required, then

another equipment rack shall be installed.

4.8.7.3. Installation Requirements

- 4.8.7.3.1. All cabling shall be installed without twists and kinks. Cables should not be looped around themselves or other objects.
- 4.8.7.3.2. Use cable management components and techniques to maintain clean, clear, and safe work environment. Do not mount cabling in locations that block access to other equipment inside and outside of equipment racks and cabinets.
- 4.8.7.3.3. Route cables with gentle loops to avoid damage due to exceeding bend radius limitations. Fiber optic cabling can be easily broken with rough handling or tight bends.
- 4.8.7.3.4. Cable slack should be concealed within the equipment racks and cabinets either vertically or within cable managers. Slack should not be looped. With the use of correct length cables, there should not be enough slack to require looping.
- 4.8.7.3.5. Patch cables should follow the side of the IT equipment rack closest to the assigned Network Interface Connection (NIC). Use correct length patch cables.
- 4.8.7.3.6. Label the cables, equipment cabinets and equipment racks as indicated in the VA [Infrastructure Standard for Telecommunications Spaces](#).

4.8.7.4. Cable Testing

4.8.7.4.1. Horizontal Cable

- 4.8.7.4.1.1. Prior to the cut-over of the equipment, test 100% of the UTP category 6A horizontal cables for performance to TIA-568-C.2, category 6A, permanent link requirements. The test instrument shall conform to the TIA-1152 Level III-e, measurement accuracy.
- 4.8.7.4.1.2. Replace and retest any cables that fail to pass the performance requirements.
- 4.8.7.4.1.3. Record the results of each test with cable identification. The test results shall be given to the VA Office of Information Technology (OIT) for each horizontal cable in electronic format.
- 4.8.7.4.1.4. The VA Project Manager shall be immediately notified if any horizontal cable fails due to link length.

4.8.7.4.2. Backbone cable

4.8.7.4.2.1. Copper cable

- 4.8.7.4.2.1.1. Prior to the cut-over of the equipment, test 100% of backbone copper cable pairs for: DC loop resistance, opens, shorts between conductors, reversed pairs, split pairs, and transposed pairs.
- 4.8.7.4.2.1.2. Replace and retest any cables that fail to pass the performance requirements.
- 4.8.7.4.2.1.3. Record the results of each test with cable identification. The test results shall be given to the VA Office of Information Technology (OIT) for each backbone cable.

4.8.7.4.2.2. Fiber Optic cable

- 4.8.7.4.2.2.1. Prior to the cut-over of equipment, test one hundred percent (100 percent) of all terminated backbone fiber strands in both directions with an Optical Power Meter and Light source to ensure the fiber strands meet or exceed the cable performance requirements of TIA/EIA-568.3-D.
- 4.8.7.4.2.2.2. Test instruments shall meet or exceed applicable requirements in TIA-568.1-D. Use only test cords and adapters that are qualified by test equipment manufacturer for channel or link test configuration.
- 4.8.7.4.2.2.3. Test multimode backbone links in both directions at

both operating wavelengths of 850 nm and 1300 nm in accordance with TIA/EIA-526-14-C, Annex A, One Cord Reference Method. The tester shall be encircled flux compliant. The Channel loss shall be 2.5 dB or less for each fiber strand.

4.8.7.4.2.2.4. Replace and retest any cables with fiber strand(s) that fail to pass the performance requirements.

4.8.7.4.2.2.5. Record the results of each test with cable identification. The test results shall be given to the VA Office of Information Technology (OIT) for each backbone cable in electronic format.

4.8.8. Special Systems

4.8.8.1. TV Distribution System

4.8.8.1.1. The Lessor will provide the following: Video cabling, pathways (conduit and cable tray), outlets, faceplates, amplifiers, splitters, backboards, cable terminations and cable testing.

4.8.8.1.2. The VA will provide the following: Video recorders, video signal processors, and Monitors.

4.8.8.1.3. A wired television distribution system connected to an antenna system or cable TV utility will be provided. Cabling will consist of 0.50" hardline or RG-11 trunk distribution cabling and RG6 horizontal cabling. Splitters and line amplifiers shall support 750 MHz minimum video bandwidth.

4.8.8.1.3.1. Splitters and amplifiers shall be located in the TR.

4.8.8.1.3.2. Locations: Waiting Rooms, Conference Rooms, Breakrooms, Police Operations.

4.8.8.2. Nurse Call System

4.8.8.2.1. Provide tone/light nurse call system, including local audible alarm, with patient stations, toilet stations, emergency call stations, staff/duty stations, master station/annunciator, dome lights and area indicators, power supplies and additional accessories as required. See Room Data Matrix for specific locations.

4.8.8.2.2. Nurse Call equipment shall be located in the TR.

4.8.8.3. Personnel Access Control System (PACS)

4.8.8.3.1. The Lessor shall design, install, and maintain an Access Control System for the facility IAW the Facility Security Level. VA Police shall have access for oversight and management.

4.8.8.3.2. Access Control equipment shall be located in the TR.

4.8.8.4. Closed Circuit Television System (CCTV)

4.8.8.4.1. The Lessor shall design, install, and maintain an CCTV system for the facility IAW the Facility Security Level and accessible remotely via the VA network.

4.8.8.4.2. CCTV equipment shall be located in the TR.

4.8.8.5. Intrusion Detection System (IDS)

4.8.8.5.1. The Lessor shall design, install, and maintain an Intrusion Detection System for the facility IAW the Facility Security Level.

4.8.8.5.2. IDS equipment shall be located in the TR.

4.8.8.6. Duress Alarm

4.8.8.6.1. The Lessor shall design, install, and maintain a Duress Alarm System for the facility IAW the Facility Security Level.

4.8.8.6.2. Duress Alarm equipment shall be located in the TR.

4.8.8.7. Public Address (PA):

4.8.8.7.1. Provide public address and mass notification (PA) system(s) covering the full VA space. Ceiling mounted speakers shall be located a maximum of 20 linear foot center to center throughout the clinic. The system shall be capable of being dialed into from any telephone for paging. System shall have the capability of paging each zone of the building separately, or to page the entire building. The minimum number

of zones shall be considered with the functional areas listed in the PFD. Review zones with the government during design. The head-end equipment for the public address system shall reside within the TR.

4.8.8.8. Intercommunication System

4.8.8.8.1. Provide intercommunications system(s) and cabling as required. Intercom system shall be located at the loading dock in view of CCTV camera. The intercom shall communicate with another intercom and be capable of dialing the front desk. The security office shall have a door release button for the loading dock entrance.

4.8.8.9. Wireless Communications

4.8.8.9.1. The VA space shall be served by two (2) separate wireless networks. The Guest Wi-Fi and VA Wi-Fi networks are separate

4.8.8.9.2. The Lessor shall provide and install the cabling infrastructure for the VA Wi-Fi network. The final location of the access points will be determined by the VA prior to the installation of said access point devices by the VA. The lessor shall provide one (1) Cat6a cable per 625 SF of ceiling space, one (1) cable in each corner (interior corners of exterior walls) and one (1) cable every 40 linear feet along the interior perimeter of the building. These cables shall be evenly spaced and distributed throughout the ceiling to provided adequate points of connections for the access points. Each cable shall be terminated in a biscuit jack. The VA may develop and provide a coverage area map noting where the Wi-Fi access points will go. In these cases, use the area map provided.

4.8.8.9.3. The Lessor shall provide Guest Wi-Fi access including installation, design, service, and operational costs. The Guest Wi-Fi system shall be designed to provide 100% coverage with established signal strength and through heat maps as identified by a wireless pre and post area coverage survey and frequency coordination study. Ensure sufficient signal strength to provide "Excellent" signal strength in the Waiting and Reception areas, and "High" signal strength throughout the rest of the Clinic Proper. Guest Wi-Fi may be unsecured and may be from common or adjacent multi-tenant space, provided the system is managed by the Lessor and is not another tenant's signal. The guest Wi-Fi system should be separate from and with no access to VA network.

4.8.8.10. Fire Alarm system. The Lessor shall install and maintain a fire alarm system in accordance with NFPA 72, latest edition. The Fire Alarm system dialer shall be on a separate phone line, external to the VA network.

5. LESSOR-PROVIDED SERVICES

- 5.1. Maintenance. General maintenance of the facility and its building service equipment. All equipment shall be maintained to provide reliable service without unusual interruption, disturbing noises, exposure to fire or safety hazards, unusual emissions of dirt. All maintenance will be done with applicable codes and manufacturer recommendations. All work done by the Lessor will be coordinated with the VA Contracting Officers Representative (COR).
- 5.1.1. Fire Suppression inspection, testing and maintenance in accordance with NFPA 25, latest edition.
- 5.1.2. Fire Alarm inspection, testing and maintenance in accordance with NFPA 72, latest edition.
- 5.1.3. Lawn and landscaping maintenance, to include any bio-pond or retention pond maintenance. Weeds shall be removed. Grass shall not exceed 3 inches in height. Leaf removal conducted to prevent unsightly appearance of the property and safe pedestrian pathways.
- 5.1.4. Replacement of exterior/interior light bulbs/lamps that are hardwired to the building power
- 5.1.5. Plumbing maintenance and repair. Drain problems that can be solved by simple application of a plunger will be handled by VA Staff. Drain problems that cannot be solved by simple application of a plunger or require repeated applications will be solved by the Lessor. All other plumbing problems are the responsibility of the Lessor and will be addressed within 24 hours of notification of Lessor's representative.
- 5.1.6. Heating, Ventilation, and Air conditioning systems maintenance and repair
- 5.1.7. Electrical maintenance
- 5.1.8. Ceiling. Lessor to replace any wet, cut, or damaged ceiling tiles or sheetrock as damage occurs. Complete ceiling tiles to be replaced and metal grid replaced or repainted every 10 years.
- 5.1.9. Painting: wall repair and touchup as needed, and wall-to-wall every three years. Vendor shall be responsible for moving and returning of furniture to allow painting. Painting shall be done after hours and must be odor free when clinic opens for business.
- 5.1.10. Window washing and window washing supplies. Windows to be washed inside and out by lessor at least every 3 months.
- 5.1.11. Exterior masonry, building structure, roof, and necessary building appurtenances to provide water tight integrity, structural soundness and acceptable appearance.
- 5.1.12. Response times. Lessor shall receive work requests during normal hours with Emergency contacts for after hours. Routine work shall be addressed within 10 business days. Emergency requests shall be responded to within two hours, and corrected within one business day or temporary measures provided.
- 5.2. Utilities. All utilities shall be provided by the Lessor unless otherwise identified in this document. This includes, but is not limited to: electric, gas, domestic water, sewer, trash removal, cable TV and Guest Wi-Fi internet.
- 5.3. Services
- 5.3.1. Snow and Ice Removal Service. Lessor shall provide snow/ice removal services for the Government on all days for which this lease has designated normal hours. Lessor shall clear parking lots, sidewalks, walkways and other entrances before accumulation exceeds 0.5 inches of snow or 0.1 inches of ice and shall provide and spread sufficient snow and ice melt product to keep the walkway, sidewalks, and parking lots from freezing over/freezing. The snow removal shall take place no later than 5:00 AM. Should accumulation continue throughout the day, the Lessor shall provide such additional snow removal services to prevent accumulation greater than the maximums specified in this paragraph. In addition to snow removal, the Lessor shall keep walkways, sidewalks and parking lots free of ice during the normal hours. The Lessor shall remove excess buildup of sand and/or ice melt to minimize slipping hazards. If the building entrance(s) has a northern exposure, then Lessor shall take additional measures to protect the safety of pedestrians.
- 5.3.2. Janitorial Services and Supplies. The Lessor is responsible for interior and exterior cleaning and refilling all paper products (paper towels, toilet paper, feminine hygiene

- products), soap dispensers and antibacterial hand sanitizer dispensers. See Appendix C for requirement details.
- 5.3.3. Recycling Services. The Lessor is responsible for establishing a recycling program for (at a minimum) paper, corrugated cardboard, glass, plastics, and metals where local markets for recovered materials exist. Where state or local law, code, or ordinance requires recycling programs, Lessor shall comply with such state and/or local law, code, or ordinance. When implementing any recycling program, the Lessor shall provide an easily accessible, appropriately sized area (2 SF per 1,000 SF of Building gross floor area) that serves the space for the collection and storage of materials for recycling. Telecom rooms are not acceptable as recycling space. Within the leased space, the Lessor shall provide the appropriate recycling containers required to facilitate and manage the recycling program for the duration of the lease.
- 5.3.4. Pest Control. The Lessor is responsible to exterminate insects, rodents and other pests monthly and when pests are detected in the leased property. Pest spraying must be done so as not to interfere with VA work. Due to sensitive nature of patients, spraying must be odorless and non-irritating.
- 5.3.5. Documentation. The lessor must provide and maintain a binder on site that includes all MSDS sheets for chemicals used for maintenance, to include any pest control products, etc.

6. GOVERNMENT-PROVIDED SERVICES

- 6.1. Business hours for patient-care are 8:00 am to 4:30 pm
- 6.2. Normal hours of occupancy are 6:00 am to 5:00 pm
- 6.3. Inspection of fire extinguishers and emergency lighting
- 6.4. Semi-annual environment of care inspection, which may result in elevated work requests for building repairs.
- 6.5. Courier Service.
- 6.6. Biohazard removal from healthcare operations.
- 6.7. Duress button testing.
- 6.8. Disposable curtain replenishment.

Appendix A: Room Data Matrix

[Appendix B: Hardware Groups and Modifiers](#)

[Appendix C: Janitorial Services](#)

Appendix A: Room Data Matrix



MadisonCo_RoomDataMatrix_251209.xls

Appendix B - HARDWARE GROUPS AND MODIFIERS

General Notes:

- A. Refer to the Room Data Matrix – Doors and Hardware for hardware group and modifier(s) assigned to each door opening. Hardware group modifiers added to numeric hardware group assignments indicate a variation to the group.
- B. Refer to Design Narrative for door hardware general notes and product information.

Notes to detailer:

- A. Add hardware group modifiers to hardware groups as required to meet program requirements.
- B. Add closers and kickplates (push side) at fire rated openings to assigned hardware group.
- C. Add overhead stops where stop condition does not allow for a wall stop.

HARDWARE MODIFIERS

- "L" Add lead-lining to all mortise hardware, including roses/escutcheons on locks. Substitute for previously specified hinges, heavy-duty pivot set and intermediate pivots spaced per manufacturers' recommendations. Lead-lined astragal at pairs is provided by door supplier. Refer to Radiation Protection specification section. Provide adequate blocking in wall for wall stops where applicable. NOTE: Installation of hardware must not compromise/penetrate lead-lining in door(s) or frame. (Use appropriate fasteners.)
- "M" Add closer holder arm at non-rated openings only.
- "P" Add door position switch/contact for doors being monitored/alarmed (both leaves of pairs). Connection by Electrical.

HARDWARE SETS

GROUP 1 – Hardware Included with Door/Frame Assembly

Door(s)/frame/ hardware is provided as part of complete assembly, unless indicated otherwise. Refer to applicable specification section(s) as indicated for door type on door detail sheet.

Also apply hardware group "modifier" when/if specifically assigned.

GROUP 11 – Privacy Function Mortise with Occupancy Indicator

Hinges as required

Spring Hinge

Note: Adjust spring hinge to partially close door without latching.

1 each Privacy Lockset w/occ. indicator

Function: Latchbolt is retracted by lever on either side unless inside thumb turn lever locks outside lever.

Operating inside lever or closing door unlocks outside lever. Outside indicator indicates occupancy when door is locked. Outside emergency release unlocks outside lever.

1 each Wall Stop

GROUP 12 – Office Function

Hinges as required

1 each Office Lockset

Function: Latchbolt is retracted by lever on either side unless outside lever is locked by inside turn button.

Key outside retracts latchbolt. Deadlocking latchbolt.

1 each Wall Stop

GROUP 13 – Classroom Function

Hinges as required

1 each Classroom Lockset

Function: Latchbolt is retracted by lever on either side unless outside lever is locked by key. Key outside locks or unlocks outside lever. Deadlocking latchbolt.

1 each Wall Stop

GROUP 14 – Storeroom Function

Hinges as required

1 each Storeroom Lockset

Function: Latchbolt is retracted by inside lever only. Outside lever is always LOCKED. Key outside retracts latchbolt. Deadlocking latchbolt.

1 each Wall Stop

GROUP 34 – Storeroom Function, with Closer

Hinges as required

1 each Storeroom Lockset

Function: Latchbolt is retracted by inside lever only. Outside lever is always LOCKED. Key outside retracts latchbolt. Deadlocking latchbolt.

1 each Closer

1 each Kickplate

1 each Wall Stop

GROUP 121 – Pushbutton Access Control (Mechanical) with Closer

Hinges as required

1 each Mechanical Pushbutton Lock (storeroom function)

Function: Latchbolt is retracted by inside lever at all times. Key outside retracts latchbolt. Outside lever is LOCKED except when valid user code is entered.

1 each Closer

1 each Kickplate

1 each Wall Stop

GROUP 200 – Electric Lock (Fail-Secure Entry with Integral Request-to-Exit Switch) Via Electronic Access Control System (Refer to Security Documents), with Closer

Hinges as required

1 each Electric Hinge

1 each Electric Lockset (fail-secure)

Function: Latchbolt is retracted by lever inside, key outside, and lever outside when unlocked electronically. Outside lever is LOCKED, unless unlocked electrically. Power off locks outside lever. Rotating inside lever connects/contacts signal switch.

1 each closer

1 each kickplate

1 each Wall Stop

1 each Door Position Switch/Contact Electronic access control system/device(s), power supplies, and monitoring/alarm(s) are provided with Security System. Contractor to coordinate the provision and installation of products. Refer to documents with Security information for location(s) and type(s) of control(s). Connection by Electrical.

OPERATIONAL DESCRIPTION: Door is normally closed and latched. Manual exit is allowed at all times by rotating inside lever, though signal will be sent. Entry is controlled by electronic access control system which secures/locks and releases/unlocks outside lever for predetermined periods of time. When secured/locked, presenting authorization temporarily unlocks outside lever to allow entry. Interruption of power secures/locks outside lever requiring key for entry. (fail-secure entry)

GROUP 850 – Interlocking Automatic Sliding Door System (with Break-Away) via Electronic Access Control System

All mechanical and electrified door hardware is provided by door assembly supplier/manufacturer.

Electronic access control system, request-to-exit device(s), and monitoring/alarm(s) are provided by Security Contractor. Refer to security documents for locations and types of controls.

Connection by Electrical.

OPERATIONAL DESCRIPTION: Doors are normally closed. Manual exit is possible at all times by “breakaway” feature on operable door panel(s). Whenever any door in the system is opened, the remaining doors are automatically and electronically locked in the closed position until the original door is again closed. Consequently, only one door of the system may be open at any one time. Interruption of power and/or fire alarm event immediately releases locks to allow passage through any door.

Appendix C. Janitorial Services

Janitorial Requirements

1. **GENERAL:** The Lessor shall provide all labor, supervision, management support, transportation, equipment, tools and materials, general supplies (replacement lights, cleaning chemicals, etc.) except Government furnished space and utilities, necessary to provide cleaning services for the Madison County VA Clinic Lease.
2. **DEFINITIONS:** As used throughout this contract, the following terms shall have the meaning set forth below:
 - a. Clean: Clean shall be defined as free of dirt, dust, spots, streaks, stains, smudges, litter, debris and other residue.
 - b. Disinfect: Cleaning in order to destroy any harmful microorganisms by application of an approved chemical agent.
 - c. Contracting Officer: The Contracting Officer is a person with the authority to enter into, administer, and/or terminate contracts and make related determinations and findings.
 - d. Lessor: The term Lessor as used herein refers to both the prime lessor and any sublessors. The prime lessor shall ensure that his/her sublessors comply with the provisions of this contract.
 - a. Contracting Officer Representative (COR): The Contracting Officer reserves the right to designate a representative to act for him/her in furnishing technical guidance and advice or generally supervise the work to be performed under this contract. Such designation will be in writing and will define the scope and limitations of the designee's authority. A copy of the designation shall be furnished the lessor.
3. **HOURS OF OPERATION:**
 - a. The Outpatient Clinic operating hours are 8:00 AM thru 4:30 PM The Lessor shall perform the services Monday through Friday, excluding Holidays between 6:00 PM – 12:00 AM or at a time established by COR. The Clinic will normally be closed on weekends and holidays.
 - b. The Lessor shall not be required to work on Federal Holidays. They are as follows and include any other days specifically declared by the President of the United States to be a national holiday: New Year's Day, Martin Luther King's Birthday, Presidents' Day, Memorial Day, Juneteenth Day, Independence Day, Labor Day, Columbus Day, Veterans Day, Thanksgiving, Christmas and any other Federal holiday declared by the President.
 - c. When a holiday falls on a Sunday, the following Monday shall be observed as a legal holiday by U.S. Government agencies. When a holiday falls on a Saturday, the preceding Friday shall be observed as a legal holiday.
4. **GOVERNMENT FURNISHED PROPERTY AND SERVICES:** In accordance with FAR 52.245-2 the Government will provide limited storage space for lessor's supplies/equipment. The use of government furnished property and services for other purposes are prohibited. Government telephone shall not be used for personal reasons.
5. **LESSOR FURNISHED ITEMS:** The lessor shall provide all facilities equipment, materials,

**Madison County VA Clinic
Agency Specific Requirements**

supplies, and service to perform the requirements of this contract. This includes replacement lights, cleaning chemicals, brushes, cloths, mops and any other supplies needed to provide the required services. Cleaning Chemicals must be approved by VA prior

**Madison County VA Clinic
Agency Specific Requirements**

to use. Lessor shall provide Safety Data Sheets (SDS) for all chemicals used. The lessor will provide paper products to include but not limited to, toilet paper, paper towel, tampon/sanitary napkin products, hand soap, tampon/sanitary napkin product bin(s) liners and trashcan liners.

6. **SCHEDULING:** Cleaning schedule will be daily Monday through Friday between 6:00 P.M. and 12:00 A.M. Jobs requiring more extensive timeframes- such as floor waxing- will be negotiated with the COR so that patient care is not interrupted

7. **STANDARDS:** Cleaning services shall meet standards established by the VA STL Health Care System in St. Louis Mo. The COR or designee for the VA Clinic will accomplish monthly inspections of the Veterans' clinic, and then issue results to the Lessor to correct any noted deficiencies. The COR will provide forms for inspections.

8. **SPECIFIC WORK REQUIREMENTS:**

- a. Lessor shall perform all tasks in accordance with contract specifications at frequency identified on the Task and Frequency Chart attached.
- b. Hazardous waste will be picked up by another lessor and is not a part of this janitorial services contract.

9. **SECURITY:** The lessor shall be issued keys for the building. The lessor will ensure that all keys issued by the Government are not duplicated. If any keys are lost or suspected lost, the lessor shall notify the COR immediately. If the keys are lost, the lessor will be responsible for the cost associated with the VA re-keying the clinic. It is the lessor's responsibility to prohibit the use of keys issued by the Government to any persons other than the lessor's employees. Only lessor employees are permitted on-site. If it is found that non-lessor individuals are on the premises, it will be grounds for immediate termination of the contract. (Reference FAR 52.212-4 (m) Termination for Cause)

10. **SPECIAL REQUIREMENTS**

a. Lessor shall be responsible for supplying, completing and submitting all reports required or requested by any Federal, State, or Local Government agency, as required by laws, ordinances, and regulations, that pertain to the supplies and services provided under this contract.

b. Lessor's employees, while performing the services of this contract, shall conduct themselves in a professional, business-like manner. The CO and COR reserve the right to restrict any Lessor employee from performing the services of this contract. The Government shall be the sole judge for this determination, which shall be followed up with written explanation to the Lessor. Smoking is permitted only in designated areas.

c. The Lessor shall, within ninety (90) minutes after receipt of notification by the COR or his/her designee, re- perform the services of this contract when it is determined that unsatisfactory performance has been received at no additional cost to the Government.

TASK AND FREQUENCY CHART

Restrooms	Daily	Weekly	Monthly	Semi- Annually	Annually	As Needed
Clean/disinfect floors	X					
Clean mirrors	X					
High dusting	X					
Clean/disinfect toilets, urinals, sinks, commodes, sanitary disposal bins.	X					
Clean/disinfect clean walls and stalls	X					

**Madison County VA Clinic
Agency Specific Requirements**

Refill dispensers (toilet paper, paper towels, soap, sanitary napkins/tampons)	X					X
Empty and replace Sanitary Disposal Bin Liners	X					
Replace string on call buttons						X

All other space	Daily	Weekly	Monthly	Semi-Annually	Annually	As Needed
VCT/Ceramic tile floors -Mop with disinfectant cleaner	X					
Clean/disinfect floors	X					
Clean cove base					X	X
Strip and wax tile floors					X	X
Top scrub and recoat				X		X
Buff tile floors	X					
High dusting			X			
High touch surfaces (doorknobs, light switches, etc.)	X					
Clean/disinfect sinks	X					
Refill dispensers (soap/paper)	X					X

Clean Glass	Daily	Weekly	Monthly	Semi-Annually	Annually	As Needed
Interior window glass			X			X
Exterior/Interior glass doors		X				X
Other exterior glass			X			X

Miscellaneous Items	Daily	Weekly	Monthly	Semi-Annually	Annually	As Needed
Clean/disinfect horizontal surfaces, including tables, cabinets, work surfaces		X				
Spot clean vertical surfaces		X				
Clean Furniture			X			X
Clean Chairs/Seating			X			X
Clean Plants/Decorative objects			X			X
Trash Cans – empty and replace liners	X					
Trash cans cleaned with disinfectant cleaner			X			
Clean/Disinfect door handles and light switches	X					
light fixtures	Remove any debris (such as insects) when present					

Note: Adequate attention must be provided to ensure all areas are maintained in a clean state.